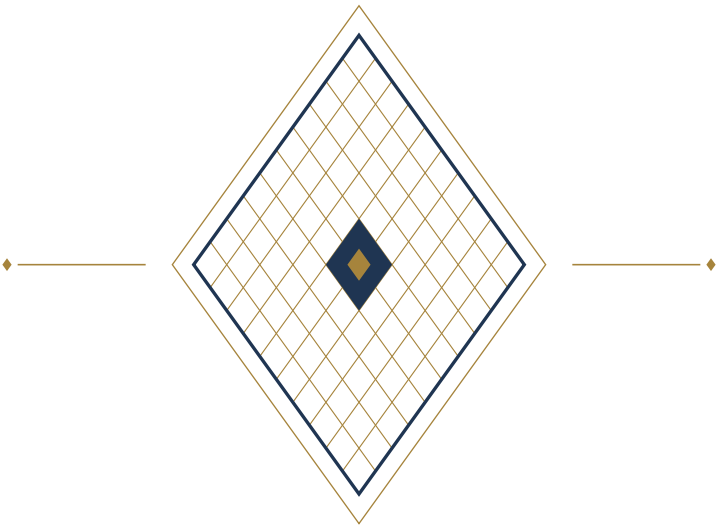
A DISSERTATION



THE SECOND PARADOX OF VALUE

Formative Scarcity, Paideiarchy, and the Uncompilable Voice in Neal Stephenson's The Diamond Age

OCTOBER 2026

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Abstract

This dissertation contends that Neal Stephenson's *The Diamond Age: Or, A Young Lady's Illustrated Primer* (1995) is misread whenever it is read primarily as a novel about nanotechnology. Its premise of molecular manufacturing functions as a controlled abolition of material scarcity, and the novel's true object is what that abolition exposes: a second compiler, the apparatus by which persons, and through persons peoples, are assembled. Taking its departure from Adam Smith's paradox of water and diamonds, the study argues that the novel stages a *Second Paradox of Value*. Once matter compilers render diamond as cheap as water, value does not disappear; it migrates toward positional goods, toward provenance, and above all toward *formative attention*, the sustained, unscalable human regard through which a child becomes a self. I call this migration the *law of displaced scarcity*, and the resource it discloses *formative scarcity*.

Four propositions follow. First, in a post-territorial order of voluntary phyles, a polity can persist only by producing persons who will choose it; the polity therefore becomes a pedagogical corporation, a regime I name *paideiarchy*. Second, Neo-Victorianism and the Confucian restoration of the Celestial Kingdom are homologous ritual technologies, mirror-image solutions to the problem of reproducing virtue once scarcity no longer disciplines conduct. Third, the eponymous Primer is not an artificial intelligence but a Mechanical Turk: its formative power depends on a hidden human performer, the ractor Miranda, whose voice exceeds the terms of her contract. That excess, the *uncompilable remainder*, is the novel's genuine diamond. Fourth, the Drummers' orgiastic collective computation and the decentralizing Seed instantiate, respectively, Georges Bataille's accursed share and Lewis Mumford's democratic technics, and both remain subordinate to the question of who authors formation.

The argument culminates in a formal proposition, *Coyote's Theorem*, named for the trickster-king who presides over the terminus of the Primer's internal romance: no apparatus of formation can compile the capacity that would exceed it. That capacity arises only where code meets an exteriority the apparatus does not script and a voice it cannot purchase. The novel's four formative trajectories (Elizabeth Finkle-McGraw, Fiona Hackworth, the Mouse Army, and Nell) compose an implicit experimental design whose outcomes (exit, vocation, allegiance, and foundation) bear the theorem out. Situating the novel among its originators (Feynman and Drexler; Plato, Rousseau, Kant, and Arendt; Mill, Arnold, and La Rochefoucauld; Confucius, Mencius, and Xunzi; Benjamin, Bourdieu, Hirschman, and Nozick) and among its fictional in-

terlocutors (Bacon, Wells, Huxley, Le Guin, Butler, Card, Chiang, Ishiguro, and Palmer), the dissertation reclassifies *The Diamond Age* as an *ethnogenetic Bildungsroman*, a narrative in which the formation of a person and the founding of a people are one event. It closes by extending the theorem to the machine tutors of the present decade and by deriving a normative corollary: a right to an uncompilable interlocutor.

Note on Text, Method, and Citation

Text. The Diamond Age is cited by part and episode rather than by page, since pagination differs between the Bantam Spectra first edition (1995) and later reprints. The novel is divided into two parts, Part the First and Part the Second, each opening with an epigraph, and its chapters carry discursive headings in the manner of eighteenth- and nineteenth-century argument-summaries rather than numbers. Every plot detail adduced here has been checked against the full text. Quotation from the novel is avoided in favour of close paraphrase; its epigraphs are cited by their sources, and the *Analects*, which the novel quotes in James Legge's translation, is cited in that translation. Stephenson's coinages (Feed, Seed, phyle, clave, thete, ractor, ractive, passive, mediatron, matter compiler) are treated as terms of art. Secondary works are cited author–date by year of first publication; canonical texts by their standard divisions (Stephanus pagination for Plato, book and passage for the *Analects* and *Mengzi*, act and scene for Shakespeare).

Method. Four procedures are combined. (1) *Genealogy*: the novel's premises are traced to their originators, with documented or nominal debts (Drexler's molecular manufacturing; Bacon's *New Atlantis*, invoked in the very name of the Neo-Victorian phyle) distinguished from structural affinities that make no claim about Stephenson's reading (Himmelfarb, Nozick, Kukathas). (2) *Philosophical criticism* in Darko Suvin's sense (1979): the matter compiler is a *novum* whose consequences the narrative works out with quasi-experimental rigour, and criticism's task is to reconstruct that experiment. (3) *Comparison by problem rather than by influence*: other speculative and philosophical fictions are adduced where they answer the same question differently. (4) *Formalization*: the final chapter restates the novel's implicit argument as axioms and a theorem, not to reduce fiction to logic but to expose the determinacy of a thought experiment the novel has already conducted.

A caution on intention. Stephenson's allusive and onomastic density rewards, and therefore tempts, over-reading. Correspondences that depend on design (the clave–*clavis* pun, the Parable of the Sower schema, the Eleusinian inversion of the climax) are flagged as speculative. The central claims are independent of intention, since they concern the logical structure of the fictional world rather than the biography of its author.

INTRODUCTION

The Diamond and the Water

0.1 *The title as an economic riddle*

In the fourth chapter of the first book of *The Wealth of Nations* (1776), Adam Smith observed that nothing is more useful than water, yet it will purchase scarcely anything, whereas a diamond has scarcely any use yet commands a great quantity of other goods in exchange. The paradox organized classical value theory until the marginalist revolution of the 1870s (Jevons 1871; Menger 1871; Walras 1874) dissolved it by relocating value from aggregate usefulness to scarcity at the margin. Stephenson's title reopens the question by abolishing its premise. In the novel's midtwenty-first century, molecular assemblers build windows, hulls, and cutlery from diamondoid lattices; the gem that the De Beers cartel and N. W. Ayer's slogan of 1947 had made the emblem of manufactured scarcity (Epstein 1982) has become a structural commonplace, delivered, like water, through municipal conduits. The Feed is a utility for matter. If diamond is now water, the novel's first and most consequential question is what has become diamond.

The title also encodes a periodization. Christian Jürgensen Thomsen's three-age system (1836) named prehistory for its dominant materials (stone, bronze, iron), a progressive sequence in which each material marks an expansion of technical power. Hesiod's *Works and Days* named the ages of man for metals in the opposite direction (gold, silver, bronze, the heroes, iron), a sequence of moral degeneration. *The Diamond Age* superimposes the two. Materially it is Thomsen's fourth age, the culmination of a progressive series. Morally, for the Neo-Victorians who dominate its foreground, it follows a Hesiodic fall, the relativist disorder of the late twentieth century against which their phyle defines itself by recoil. The novel's governing dialectic is present in its first two words: material ascent and moral descent, and the question whether formation can reverse the second without forfeiting the first.

The novel announces this dialectic in its paratext. Its epigraph is from Confucius, in Legge's translation: 'By nature, men are nearly alike; by practice, they get to be wide apart' (*Analects* 17.2). It is a one-line statement of the primacy of formation over endowment. Part the First opens with Sir Charles Petrie's account, in *The Victorians* (1960), of moral reform and decline as a pendulum driven by reaction against the habits of the preceding period rather than by a few distinguished individuals; it is the Neo-Victorians' history of themselves. Part the Second opens with Qiying, the Qing commissioner who negotiated the Treaty of Nanking in 1842, observing

that barbarians born beyond the empire continually misconstrue the administration of the Celestial Dynasty. Formation, reaction, and the opium settlement: the three epigraphs name, in order, the three strata this dissertation excavates.

0.2 *The argument*

The novel interleaves two plots that criticism has tended to treat as unequal partners. In the domestic plot a stolen interactive book raises a destitute girl; in the geopolitical plot a centralized matter-distribution grid (the Feed) is contested by a decentralized generative technology (the Seed) amid a neo-Boxer insurrection in China. This dissertation argues that the two plots are one. Both concern the conditions under which something can be reproduced without being controlled, and both resolve when the reproducible meets what cannot be reproduced.

The thesis may be stated in a sentence. *When the scarcity of matter is abolished, the scarcity of formative attention becomes the master scarcity from which every political form in the diamond age derives, and the novel's deepest claim is that this scarcity cannot be engineered away, because its value consists precisely in its exceeding every specification.* Four propositions elaborate it.

1. *Displaced scarcity.* Abundance does not abolish scarcity but relocates it: to positional goods (Hirsch 1976), to provenance and aura (Benjamin 1935–36), and to the attention by which persons are formed (Simon 1971). The last yields the Second Paradox of Value: the good on which all valuation depends is the good valuation cannot price.
2. *Paideiarchy.* In a world of universal exit and irrelevant territory, a polity persists only by producing members who will choose it. Sovereignty therefore migrates from the control of land to the authorship of formation. The phyle is not a nation that keeps a school but a school that keeps a nation.
3. *The uncompilable remainder.* The Primer's formative power derives not from its code, which Dr. X replicates by the hundred thousand, but from Miranda, the human performer concealed within it, whose regard exceeds her wage. The remainder is the irreducible element without which formation yields allegiance but not autonomy.
4. *Coyote's Theorem.* No apparatus of formation can compile the capacity that would exceed it. Sovereignty with respect to a formative system arises only from the conjunction of that system with an exteriority it does not script and a voice it cannot purchase.

0.3 *Situating the intervention*

The novel's reception has followed three tracks. The first reads it as postcyberpunk extrapolation, in the sense Lawrence Person gave that term (1998), and measures it against the Drexlerian program and its critics. The second reads its phyles and its Neo-Victorian apologetics as a contribution, admiring or suspicious, to the libertarian and neoconservative social theory of the 1990s. The third, chiefly a phenomenon of the past two decades, treats the Primer as a design brief: the figure to which educational technologists and AI laboratories return when they describe the adaptive tutor they mean to build, a reception Stephenson himself theorized when he called on science fiction to supply engineers with motivating images (Stephenson 2011; Finn and Cramer 2014). Each track isolates one stratum. The present study subordinates all three to a single problem, the political economy of formation, and argues that only from that vantage do the novel's nanotechnology, its sociology of tribes, and its pedagogy appear as parts of one argument.

0.4 *Plan of the work*

Chapter 1 reconstructs the technological premise from Feynman to Drexler, reading the Feed and the Seed through Mumford's distinction between authoritarian and democratic technics and through the nineteenth-century history of opium and treaty ports that the novel rewrites. Chapter 2 develops the Second Paradox of Value. Chapter 3 analyzes the phyle as a post-territorial political form, from Cleisthenes to Nozick and Kukathas, and introduces paideiarchy. Chapter 4 examines Neo-Victorianism as moral technology, centring Finkle-McGraw's apology for hypocrisy. Chapter 5 reads the Celestial Kingdom as Neo-Victorianism's Confucian mirror. Chapter 6 places the Primer in the history of the philosophy of education, from Plato's *Republic* to Papert's *Mindstorms*. Chapter 7 analyzes the ractor and the remainder. Chapter 8 treats the Drummers, the Seed, and CryptNet as the return of Bataille's accursed share. Chapter 9 sets the novel within a comparative constellation of speculative and philosophical fiction. Chapter 10 states and defends Coyote's Theorem and draws its consequences for the machine pedagogies of the present.

CHAPTER

I

Compilers

The Technological Premise and Its Genealogy

The matter compiler is the novel's *novum*, and its political meaning lies less in what it makes than in how its inputs are routed. This chapter traces the device to its scientific originators, then reads the two infrastructures built upon it, the Feed and the Seed, as competing constitutions of material life.

1.1 *From Feynman's room to Drexler's assembler*

The lineage begins with Richard Feynman's 1959 address to the American Physical Society, "There's Plenty of Room at the Bottom" (published 1960), which proposed that nothing in physical law forbids the manipulation of matter atom by atom. K. Eric Drexler converted the conjecture into a research program. *Engines of Creation* (1986) introduced the molecular assembler, the self-replicating device, and the spectre of runaway replication later popularized as 'grey goo'; *Nanosystems* (1992) supplied the engineering apparatus, including diamondoid mechanosynthesis and mechanical 'rod logic' computation. Stephenson's world is Drexlerian in its particulars as well as its premise, and the novel's acknowledgments thank Drexler by name. Its computation is mechanical (one engineering works keeps five cubic centimetres of rod logic in a cabinet, drawing some hundred thousand watts through the Feed's superconducting lines), its structural material is diamondoid, and its domestic compilers assemble goods from molecular feedstock. Robert Freitas and Ralph Merkle later systematized the field of kinematic self-replicating machines (Freitas and Merkle 2004), whose conceptual ancestor is John von Neumann's universal constructor (von Neumann 1966).

The premise was contested. Richard Smalley's objections (2001), formalized in his exchange with Drexler in *Chemical & Engineering News* (2003), held that 'fat' and 'sticky' manipulator fingers make positional assembly chemically impossible. The dispute is instructive precisely because it is irrelevant to the novel's argument. Stephenson requires not that Drexlerian assembly be feasible but that material scarcity be hypothetically abolished, so that whatever remains scarce can be observed in isolation. The compiler is an experimental control.

1.2 *Infrastructure as constitution: the Feed*

Abundance in the novel is not ambient but plumbed. Raw elements are processed at Sources and delivered through Feed lines to compilers in homes and public places, where licensed designs are executed. Three features make the Feed a constitution rather than a utility. It is *centralized*, so that whoever holds the valve holds the population. It is *tethered*, so that compilers act only on licensed designs and leave a trace, which is why John Percival Hackworth must take his illicit copy of the Primer to Dr. X's unmonitored compiler. And it is *phyletically owned*, so that the distribution of matter tracks the distribution of cultural power.

Lewis Mumford's distinction between authoritarian and democratic technics (1964), between system-centred apparatus of immense power and person-centred technique of modest scale, describes the Feed with precision. Langdon Winner's demonstration that artifacts embody political arrangements (1980), Lawrence Lessig's thesis that architecture regulates as law does (1999), and Jonathan Zittrain's opposition between generative platforms and tethered appliances (2008) each restate, in a later idiom, what the novel dramatizes: the Feed governs by conduit. Its nineteenth-century prototype is Edward Bellamy's *Looking Backward* (1888), whose centralized distribution system made abundance a function of administrative unity. Its twentieth-century theory is Gilles Deleuze's society of control (1990), in which power no longer encloses bodies but modulates flows.

1.3 *The Seed: generativity, and a rhizome shaped like a tree*

The Seed is the Feed's negation: a self-contained generative package that grows its products from local materials without conduits, licences, or a central Source. Its technical genealogy runs through von Neumann's self-reproducing automaton, but its political ancestry is older and agrarian. Gandhi's *Hind Swaraj* (1909) made the spinning wheel the instrument of a sovereignty founded on decentralized production; E. F. Schumacher's *Small Is Beautiful* (1973) theorized appropriate technology; Ivan Illich's *Tools for Conviviality* (1973) distinguished tools that enlarge autonomy from those that enforce dependence; and William Morris's *News from Nowhere* (1890), written explicitly against Bellamy, imagined abundance without centralization. The Feed–Seed opposition replays the Bellamy–Morris debate in a molecular register.

Deleuze and Guattari's *A Thousand Plateaus* (1980) opposed the arborescent model of power, hierarchical and rooted in a single trunk, to the rhizome, acentred and connective. The novel inverts their iconography with some wit. The Feed, a network, functions as a tree: every branch returns to a root at the Source. The Seed, which literalizes the tree, functions as a rhizome: every seed is its own root, and the whole has no centre. The inversion warns against equating decentralization with liberation. In the novel the Seed is coveted by CryptNet's crypto-anarchists and by the Confucian restorationists of the Celestial Kingdom alike, for opposite ends. Generativity is politically indeterminate. It relocates the question of sovereignty without answering it.

1.4 *Toner, mites, and the ecology of abundance*

Abundance has exhaust. The air of the novel's cities is thick with microscopic devices, defensive and predatory, and with the inert residue of their engagements, colloquially 'toner'. Wealthy claves are protected by aerostatic immune systems that screen what crosses their perimeters; the thetes breathe what is screened out, and the asthma that hospitalizes Nell's brother Harv is the bodily signature of the arrangement. Ulrich Beck's *Risk Society* (1986) argued that advanced modernity shifts distributive conflict from goods to 'bads', from the allocation of wealth to the allocation of hazard. The diamond age completes Beck's transition: goods are nearly free, and stratification is expressed as the differential exposure of lungs. Donna Haraway's analysis of immune-system discourse as a biopolitics of self and other (1989) explains why the clave's defensive grid is figured as an immune system. The enclave is imagined as a body whose health consists in exclusion.

1.5 *The opium homology*

The novel's Chinese geography is a palimpsest of the nineteenth century, and the homology holds only if that century's agents are named. British merchant houses, Jardine, Matheson & Co. foremost among them, smuggled opium grown under the East India Company's Bengal monopoly into China in defiance of Qing prohibition. In 1839 Commissioner Lin Zexu confiscated and destroyed some 20,000 chests at Humen and wrote to Queen Victoria protesting the poison her subjects sold; Britain answered with an expeditionary force. The Treaty of Nanking (1842) ceded Hong Kong Island, opened five treaty ports, and imposed an indemnity. In the second war (1856–60) Britain and France compelled the legalization of the trade, and British and French troops looted and burned the Old Summer Palace on Lord Elgin's order. The Leased Territories recall the treaty ports and extraterritorial concessions those wars secured, the Shanghai International Settlement above all, and New Chusan, the artificial island on which the Neo-Victorians of Atlantis/Shanghai reside, bears the name of the island the British occupied in 1840. The Feed occupies the structural place of opium in a precise sense: not merely a cheap import that dissolves social cohesion, but a dependency secured by force and protected by extraterritorial law. The moral reasoning of the Celestial Kingdom echoes Lin's. The Fists of Righteous Harmony re-enact the Yihetuan of 1899–1901, who tore up railways and telegraph lines as conduits of foreign power and killed tens of thousands of Chinese Christians and some two hundred foreign missionaries (Section 5.5; Esherick 1987; Cohen 1997); their successors in the novel turn on the Feed.

The homology is the novel's own. Late in the book Dr. X tells Hackworth that since the Opium Wars China has struggled to absorb the *yong*, the outward function, of Western technology without its *ti*, its underlying essence; that just as the ancestors could not open the ports without accepting the poison of opium, China could not open itself to Western technology without importing Western ideas; and that the result has been centuries of chaos, which the Seed is to end. His vocabulary is the late-Qing reformers' formula, 'Chinese learning as essence, Western learn-

ing as function', popularized by Zhang Zhidong's *Exhortation to Learning* (*Quanxue pian*, 1898), a treatise whose title echoes Xunzi's essay on learning (Section 5.2). An early passage of narration supplies the counter-image. Beside new buildings grown with their Feed lines built in like capillaries, the old neighbourhoods of Shanghai, without Feed or fed by lines propped on bamboo stilts, look as inert as an opium smoker lost in an ancient dream amid the bustle of a modern street. The modern eye sees the unplumbed city as the addict. Dr. X sees the plumbing as the drug.

The homology clarifies the stakes of the Seed. It is the anti-opium, a technology of material autarky that would end the century of dependence the Feed reinstates. It also marks the limit of material solutions. China's crisis, as the novel construes it, is a crisis not only of supply but of formation: abandoned children, degraded traditions, and anomic coastal cities. The Seed can feed such a population. It cannot raise one. The remainder of this dissertation concerns the apparatus the novel sets beside the matter compiler to answer that second need.

CHAPTER

II

The Second Paradox of Value

Political Economy after Matter

THE DIAMOND AGE does not end economics; it relocates the economic problem to the goods that compilation cannot touch. This chapter identifies three such classes (positional goods, aura-bearing goods, and formative attention) and argues that the third differs from the other two in kind.

2.1 *From the paradox of value to the margin of zero*

Smith's paradox survived Ricardo's labour theory of value and fell to the marginalists, who showed that exchange-value tracks the utility of the last available unit, so that abundant water is cheap and scarce diamonds are dear. In the diamond age the marginal cost of compiled goods approaches the cost of feedstock, energy, and licensed design. Jeremy Rifkin's 'zero marginal cost society' (2014) names the limit; Marx's 'Fragment on Machines' in the *Grundrisse* (1857–58) had anticipated its consequence. When production rests on the general intellect embodied in machinery rather than on immediate labour-time, labour-time ceases to measure wealth, and capital preserves its form by enclosing knowledge. The novel bears Marx out. Its residual economy of compiled goods is an economy of rents on designs, which is why the Primer's original belongs to an Equity Lord, and why Hackworth's crime is, at bottom, an infringement of intellectual property.

2.2 *Positional goods and the economy of distinction*

Fred Hirsch's *Social Limits to Growth* (1976) defined positional goods as those whose value depends on relative standing and whose scarcity is therefore social rather than technical: no increase in productivity enlarges the supply of first places. Thorstein Veblen's *Theory of the Leisure Class* (1899) had shown how conspicuous consumption and leisure convert expenditure into rank. Pierre Bourdieu's *Distinction* (1979) showed how taste, manners, and educated dispositions function as cultural capital, convertible into economic and social capital and transmitted through the embodied habitus.

The Neo-Victorian phyle is an apparatus for producing positional goods in a world without material ones. Its peerage of Equity Lords, its paper books and hand-sewn garments, the artisan community of Dovetail with its premium on handmade objects, the admission lists of Miss Matheson's Academy, and membership itself are goods whose value derives from exclusion. Walter Benjamin's essay on technological reproducibility (1935–36) explains the mechanism: reproduction withers the aura of the work, its unique existence in time and place, and in an order of total reproducibility the unreproduced acquires a correspondingly intensified aura. Dovetail's craftsmen extend John Ruskin's defence of the dignified imperfection of handwork in 'The Nature of Gothic' (1853), and Morris's Arts and Crafts movement, into a world in which the machine can imitate every surface of the handmade except its provenance. Jean Baudrillard's distinction between functional value and sign-value (1968) completes the picture: once function is free, objects circulate as signs of the persons who own them.

2.3 *Keynes's grandchildren*

In 'Economic Possibilities for our Grandchildren' (1930), John Maynard Keynes predicted that the economic problem would be solved within a century and that humanity would then confront its permanent problem: how to occupy its freedom from pressing economic cares so as to live wisely, agreeably, and well. He feared a general nervous breakdown among those unprepared for leisure. The thetes are Keynes's grandchildren in the worst case. Public compilers furnish subsistence; interactive and passive entertainments fill the hours that labour once occupied; the household into which Nell is born, with its succession of violent men and a mother absorbed in mediatronic diversion, is subsistence without formation.

The Victorian tradition the Neo-Victorians invoke had names for this condition. Disraeli's *Sybil* (1845) diagnosed England as 'two nations' between whom there was no intercourse and no sympathy. Matthew Arnold's *Culture and Anarchy* (1869) sorted society into Barbarians, Philistines, and Populace, and proposed culture as the remedy for anarchy. Marx's portrait of the lumpenproletariat in *The Eighteenth Brumaire* (1852), a population available to any adventurer, anticipates the thetes' availability to the Fists. Neil Postman's argument in *Amusing Ourselves to Death* (1985), that the epistemology of entertainment media corrodes the capacities print literacy cultivated, supplies the novel's media theory. Among the Primer's decisive acts is to teach its owner to read for herself rather than be read to.

The novel's sympathies here are not innocent. Its portrait of thete life shades toward the 'culture of poverty' thesis associated with Oscar Lewis (1959; 1966), in which the poor are made to bear responsibility for dispositions their conditions produce. Chapter 4 argues that the plot quietly refutes this reading, since Nell rises not through the self-reform of thete culture but through inputs from beyond it.

2.4 *Attention, care, and the law of displaced scarcity*

Herbert Simon observed (1971) that a wealth of information creates a poverty of attention. The diamond age generalizes the insight: a wealth of matter creates a poverty of formation. Formative attention has four properties that distinguish it from positional and aura-bearing goods. It is *time-indexed*, consumed in the moment of its provision and incapable of being stockpiled. It is *person-indexed*, addressed to a particular child whose responses alter what is given. It is *non-scalable*, because its output is the labour-time itself. And it is *non-contractual in its decisive increment*, since what forms a child is the regard given beyond what any arrangement requires.

The third property has an exact economic description. William Baumol and William Bowen (1966) identified a 'cost disease' afflicting the performing arts: a string quartet requires the same four players and the same minutes it required two centuries ago, so that its relative cost rises as productivity improves elsewhere; Baumol later extended the diagnosis to education and care. The ractor, a live performer whose performance is a lesson, sits at the intersection of the sectors Baumol judged incurable. The fourth property has a feminist genealogy. Silvia Federici's *Wages Against Housework* (1975) argued that the unwaged labour of social reproduction underwrites all waged labour; Arlie Hochschild's *The Managed Heart* (1983) anatomized emotional labour as a commodity; Nancy Fraser (2016) diagnosed a structural 'crisis of care' in which capital depletes the reproductive capacities on which it depends.

The Second Paradox of Value can now be stated. In the diamond age, compiled matter reproduces the position of Smith's water: indispensable, abundant, cheap. Positional goods reproduce the position of Smith's diamond: scarce, dear, of little use. Formative attention occupies a third position that Smith's dyad could not anticipate. It is indispensable to the production of the very subjects for whom anything has value, yet it is systematically underpriced, because its value consists in its exceeding exchange. *The good on which all valuation depends is the one valuation cannot price.*

From this follows the *law of displaced scarcity*: when a domain becomes abundant, scarcity relocates to the conditions under which that abundance can be meaningfully used. In an economy of matter, the scarce condition is a formed self. Every institution in the novel (the phyle, the Academy, the Primer, the Mouse Army, the Drummers) is a strategy for producing, capturing, industrializing, or dissolving that condition.

CHAPTER

III

Phyle, Clave, Thete

The Political Philosophy of Post-Territorial Order

THE PHYLE IS the novel's answer to the decay of the territorial state, and its political theory is more exact than its genial surface suggests. This chapter recovers the Athenian lexicon beneath Stephenson's coinages, places the phyle within modern theories of elective community, and argues that its logic of persistence turns every polity into a school.

3.1 *An Athenian lexicon*

Stephenson's vocabulary is drawn from Athenian constitutional history, and the borrowing is exact. The *phylē* was a tribal division of the citizen body. Cleisthenes' reforms of 508/7 BCE replaced the four ancestral Ionian tribes with ten new ones, each composed of *trittyes* drawn from the city, the coast, and the interior, in order to break the regional and kinship networks of aristocratic power (Aristotle, *Athenaiōn Politeia* 21). The Cleisthenic tribes were, in the novel's idiom, synthetic phyles: artificial solidarities engineered to supersede inherited ones. The *thētes* were the lowest of Solon's four census classes, men without the property to equip themselves as hoplites. Their service as rowers in the trireme fleet gave them leverage that underwrote radical democracy, a connection the anonymous 'Old Oligarch' observed with disapproval: it was just, he conceded, that the poor should have more say, since it is they who row the ships and give the city its power (Pseudo-Xenophon, *Athenaiōn Politeia* 1.2).

The diamond age's thetes have no fleet. They are the propertyless without leverage, citizens of nothing. The novel's climax restores their Athenian function in an unexpected form, when a host composed of the abandoned becomes the instrument of a new polity. A third coinage, *clave*, derives from *enclave* and *conclave*, both rooted in Latin *clavis*, 'key'. A clave is a keyed space. Whether by design or by the logic of its material, the Primer's romance is organized around Princess Nell's quest for twelve keys, taken from castle after castle, and Nell's progress through the novel is a passage through successively locked enclosures: from her mother's flat to Dovetail, to the Academy, into the Coastal Republic, and finally beneath the sea.

3.2 *From imagined to elective community*

Benedict Anderson's *Imagined Communities* (1983) explained the nation as a community imagined through the shared temporality of print-capitalism. The phyle is an imagined community whose medium is the network rather than the newspaper and whose membership is chosen rather than inherited. Joel Kotkin's *Tribes* (1993) argued that success in the global economy would accrue to dispersed 'global tribes' bound by identity, trust, and an ethic of self-improvement, and named five: the Jews, the British, the Japanese, the Chinese, and the Indians. The novel's dominant phyles (the Neo-Victorians of New Atlantis, the Nipponese, the Han, and the looser agglomeration of Hindustan) reproduce four of Kotkin's five. The correspondence is close enough to suggest, if not to establish, direct debt.

Max Weber's distinction between class and status (*Economy and Society*, 1922) supplies the sociological category. Classes are defined by market situation; status groups (*Stände*) by a style of life, a claim to honour, and restrictions on commensality and marriage. Phyles are Weberian status groups detached from both class and territory. H. G. Wells supplied the literary prototype in *A Modern Utopia* (1905), whose ruling 'samurai' form a voluntary order bound by a common Rule and open to anyone who will adopt its discipline. The Neo-Victorians, who admit members of any ancestry prepared to take their Oath, as Hackworth did at twenty-one, are Wells's samurai in Victorian dress.

3.3 *The framework for utopia and its discontents*

The phyle system realizes with unusual fidelity the 'framework for utopia' of Robert Nozick's *Anarchy, State, and Utopia* (1974): a meta-utopia within which many communities, each pursuing its own conception of the good, coexist under a minimal order that guarantees the right of exit. Its economics are Charles Tiebout's (1956), in which citizens sort themselves among jurisdictions by voting with their feet. Its law is the polycentric order David Friedman sketched in *The Machinery of Freedom* (1973). Its political theory is Chandran Kukathas's *The Liberal Archipelago* (2003), in which the only universal right is freedom of association and exit, and John Gray's modus vivendi liberalism (2000), which seeks not consensus on justice but terms of peaceful coexistence among irreconcilable ways of life. The Common Economic Protocol, the thin treaty governing commerce among phyles, is a modus vivendi in this precise sense: an agreement about trade, not about right.

The novel shows the Protocol at work in its opening chapters. After Bud fires his skull gun at a stranger travelling with his wife and daughter, a party of Ashanti who have been tracking him wrap him in shrinking film at the Aerodrome, and a mounted man places him under 'personal arrest' for violating the Common Economic Protocol. The arrest is a private act; the trial is not. Because the territory belongs to a nation-state that recognizes the Protocol, Bud is entitled to a hearing in that state's courts, here those of the Chinese Coastal Republic. Enforcement is polycentric and jurisdiction territorial: the phyles police the streets, and the residual nation-state sup-

plies the court. For a thete like Bud, that court is the only political body that will ever take notice of him.

Albert Hirschman's *Exit, Voice, and Loyalty* (1970) predicts the system's characteristic pathology. Where exit is cheap, voice atrophies, organizations decline without the correction voice supplies, and loyalty becomes the scarce currency that phyles must manufacture. John Rawls's *Political Liberalism* (1993) predicts its instability: a mere modus vivendi endures only while the balance of power holds, because no party affirms its terms for their own sake. The Fists' uprising tests both predictions. It erupts among populations the Protocol does not represent, and it succeeds because the Protocol commands no loyalty.

3.4 *The thete and the right to have rights*

Hannah Arendt's analysis of statelessness in *The Origins of Totalitarianism* (1951) is the indispensable lens for the thete, and its history belongs in the argument. Arendt wrote out of the interwar refugee crises and the Nazi regime's denationalization of German Jews, begun by the Reich Citizenship Law of 1935 and completed in 1941 by a decree stripping citizenship from Jews deported abroad, a legal erasure that preceded their murder; she was herself stateless from 1937 to 1951. The stateless, she argued, lose not this or that right but the right to have rights: membership in a community willing and able to guarantee any rights at all. The thetes live under the commercial regime of the Protocol and the jurisdiction of whatever authority claims the ground beneath them, but they belong to no phyle that will act on their behalf, and the novel shows what this absence permits. Burt, the man their mother Tequila takes in, beats Nell, Harv, and Tequila herself. A man with a record of such offences is suspected, by Judge Fang's own constable, of having molested Nell; when Harv and his friends beat that man, it is Harv whom the court canes. The state meets these children only as punishment, and their one remedy is flight.

Giorgio Agamben's *homo sacer* (1995), the figure who may be killed without the killing counting as either sacrifice or crime, which Agamben traced from archaic Roman law to the camp, finds an exact counterpart in the novel's opening. Bud, Tequila's lover and Harv's father, is held after his arrest in an open-air prison on the Yangtze delta whose perimeter is enforced by devices mounted on the prisoners' bones: crossing the line kills, corpses lie just beyond it, and the inmates lynch a suspected thief by hurling him across. Inside that enclosure killing carries no legal consequence; it is Agamben's camp. Bud does not die there. Judge Fang, finding no mitigating circumstances, sentences him to death and sends him out to stand at the end of a pier on the Huangpu, and the court's records later note simply that he was executed. The thete is not killed outside the law. He meets the law only as punishment, never as protection, and passes on his way through a space from which the law has withdrawn.

James C. Scott's *The Art of Not Being Governed* (2009) describes peoples who flee the state by design. The thetes are not such people. They have not escaped the phyles; they have been left outside them. Kukathas's archipelago has no answer for those who cannot exit because they never entered. This is the deepest objection to the phyle order, and the novel registers it: the framework for utopia is a utopia only for those already formed to choose.

3.5 *Heterotopias and their afterlives*

Michel Foucault's lecture 'Of Other Spaces' (1967) named heterotopias: real places in which the other sites of a culture are represented, contested, and inverted. Samuel R. Delany's *Triton* (1976), subtitled *An Ambiguous Heterotopia* in answer to the subtitle of Ursula K. Le Guin's *The Dispossessed* (1974), *An Ambiguous Utopia*, imagined a society of chosen subcultures on a moon of Neptune. China Miéville's *The City & the City* (2009), in which two polities occupy the same ground and their citizens are trained to 'unsee' one another, supplies a phenomenology of the clave that the novel implies but never states. Stephenson's own *Snow Crash* (1992), with its franchise-organized quasi-national entities and walled 'burbclaves', is the commercial prototype that *The Diamond Age* moralizes: sovereignty there is a franchise; here it is a culture.

The phyle has since migrated from fiction into programme. Patri Friedman's seasteading movement (2008), Paul Romer's proposals for charter cities (2009), and Balaji Srinivasan's *The Network State* (2022) each describe elective, non-territorial or extraterritorial polities in terms the novel anticipated. They tend also to reproduce its blind spot. They are designed for those who will choose them and say little about those not equipped to choose.

3.6 *Paideiarchy: the phyle as a school with an army*

The decisive consequence of the phyle system follows from two of its features: exit is universal and territory is irrelevant. A territorial state can rely on birth and borders to supply its citizens; a phyle can rely on neither. Its children may leave, and many will be tempted. The only means by which a phyle can persist across generations is to produce persons who will choose it, and the only institution capable of that is formation. In such an order the decisive organ of sovereignty is neither the court nor the army but the school, together with the stories by which children come to love what they will later be asked to choose.

The point has an ancient pedigree. Plato's *Republic* demonstrates, among other things, that a just city is first an educational institution; Werner Jaeger's *Paideia* (1934–47) reconstructed Greek culture as a sustained reflection on the formation of the human being. Friedrich Hayek's *The Fatal Conceit* (1988) argued that moral traditions spread because the groups that adopt them prosper; the Neo-Victorian justification of Victorian morals is Hayekian in exactly this sense. Alasdair MacIntyre's *After Virtue* (1981) argued that the virtues are intelligible only within traditions embodied in communities and practices, and closed by awaiting a new St Benedict to found such communities amid the new dark ages. The phyles are MacIntyrean communities founded on a market.

I propose the term *paideiarchy* for a regime in which sovereignty is exercised principally through the authorship of formation. The phyle is paideiarchic by necessity. It is not a nation that keeps a school but a school that keeps a nation, an inversion of the remark attributed to Mirabeau that Prussia was not a state with an army but an army with a state. The geopolitics of the novel follow. The 1990s debate between Francis Fukuyama's end of history (1989; 1992) and Samuel Huntington's clash of civilizations (1993) presumed, respectively, a single victorious civi-

lization and several warring ones. The diamond age proposes a third model: a market in civilizations, which compete for members as firms compete for customers, and which compete above all by pedagogy. In such a market the Primer is the most strategic object in the world.

CHAPTER

IV

The Apology for Hypocrisy

Neo-Victorianism as Moral Technology

NEO-VICTORIANISM is not nostalgia but engineering: the deliberate adoption of a moral code for its reproductive efficacy. This chapter reconstructs its central argument, Lord Alexander Chung-Sik Finkle-McGraw's defence of hypocrisy, traces its sources, and identifies the founder's paradox that motivates the commission of the Primer.

4.1 *Finkle-McGraw's disquisition*

In the course of deciding how to treat Hackworth's transgression, Finkle-McGraw offers an account of why his generation rehabilitated hypocrisy. In the late twentieth century, he argues, moral relativism disabled criticism: if there are no standards, no conduct can be condemned by reference to them. Yet the appetite for censure did not abate. It found the one vice relativism could still prosecute, the gap between what one professes and what one does, and elevated this commonplace frailty into the gravest of sins. The result was perverse. The surest way to escape the charge of hypocrisy was to profess nothing. The Neo-Victorians reverse the valuation. To fall short of an ideal does not impeach the ideal; the man who fails a standard he affirms has at least affirmed it.

The disquisition is staged over lunch in a pub, where Finkle-McGraw puts the question of hypocrisy to Hackworth in the presence of Major Napier. He adds that in his youth to call someone a Victorian was almost to call him a fascist, so that the rehabilitation of hypocrisy and the rehabilitation of the Victorians are a single gesture. The Confucian strand supplies a counterpart. Weighing a case of baby trafficking, Judge Fang reflects that the Coastal Republic believes in no virtue at all and so cannot even aspire to hypocrisy. La Rochefoucauld's maxim is affirmed from both sides of the novel's mirror: hypocrisy presupposes a standard, and only an order with standards can fall short of them.

The argument has a distinguished ancestry. La Rochefoucauld's maxim that hypocrisy is the homage vice pays to virtue (*Maximes*, 1665, no. 218) states its premise. Judith Shklar's chapter 'Let Us Not Be Hypocritical' in *Ordinary Vices* (1984) supplies its liberal analogue: the zeal to unmask hypocrisy, Shklar argued, is itself a vice in a liberal society, cruel in its methods and cor-

rosive of the civility pluralism requires. David Runciman's *Political Hypocrisy* (2008) extended the case. Lionel Trilling's *Sincerity and Authenticity* (1972) supplies the historical frame. The Neo-Victorian order restores what Trilling called sincerity, the congruence of avowal and conduct within a public role, against authenticity, fidelity to an inner self that disdains roles. That the novel's emblematic profession is the ractor, a performer whose truthfulness consists in the role, is not accidental.

4.2 *Manners as infrastructure*

Norbert Elias's *The Civilizing Process* (1939) traced how external constraints on conduct are internalized as self-constraint, so that manners become a second nature through which social interdependence is managed. Erving Goffman's *The Presentation of Self in Everyday Life* (1956) described social life as dramaturgy, with front and back regions. Neo-Victorian etiquette is Eliasian self-constraint deliberately reinstalled and Goffmanian performance raised to a principle of state.

Walter Bagehot's *The English Constitution* (1867) distinguished the 'dignified' parts of government, which excite reverence and secure allegiance, from the 'efficient' parts, which actually rule. New Atlantis has the same bicameral soul. Its monarchy, its peerage of Equity Lords, and its ceremonial life are dignified; Machine-Phase Systems, the Feed, and the security apparatus that employs Napier are efficient. The novel's insight is that in post-scarcity the dignified part becomes the more important, since it alone generates the loyalty that exit-rich populations otherwise withhold.

The phyle's ethic of improvement descends from Samuel Smiles's *Self-Help* (1859), the Victorian gospel of perseverance. Nell's ascent resembles a Smilesian parable and is not one. It depends on a theft, a magistrate's discretion, a stranger's devotion, and an engineer's design. The novel stages a Smilesian plot in order to show that self-help is always help.

4.3 *Two Atlantises*

The name of the Neo-Victorian phyle invokes Francis Bacon's *New Atlantis* (1626), whose island of Bensalem is guided by Salomon's House, a college of natural philosophers whose research is an instrument of statecraft and whose piety is undisturbed by its power. The fusion of Anglican propriety with molecular engineering in Stephenson's New Atlantis is Baconian in precisely this sense. Bacon's program, the advancement of knowledge for the relief of man's estate, here reaches completion, and the Neo-Victorian problem is what to do with it.

The name also invokes the older Atlantis of Plato's *Timaeus* (24e–25d) and *Critias*, a thalassocracy whose hubris was punished by submersion. The warning is not idle. Beneath the dignified surface of the diamond age lies the sea, and in it the Drummers, whose submerged collectivity is the order's unacknowledged substrate (Chapter 8).

4.4 *The Victorian double*

The historical Victorians were not as their apologists imagine. Steven Marcus's *The Other Victorians* (1966) documented the period's pornographic underworld, and Foucault's *The History of Sexuality* (1976) dismantled the 'repressive hypothesis': the nineteenth century did not silence sexuality but multiplied the discourses that produced it. Robert Louis Stevenson's *Strange Case of Dr Jekyll and Mr Hyde* (1886) gave the double life its emblem, and Wilde's *The Importance of Being Earnest* (1895) made it a comic technology, 'Bunburying', by which respectability is maintained through an invented elsewhere.

Stephenson's Neo-Victorians reproduce the structure along with the manners. The erotic ractive houses of the Coastal Republic, in one of which Nell later performs, and the orgiastic depths of the Drummers, into which Hackworth disappears for a decade, are not contradictions of the Neo-Victorian order but its back region. Hackworth, a respectable artifex who submerges into the collective, is Jekyll in a frock coat. The apology for hypocrisy is therefore also an apology for the underside: a society that accepts the gap between profession and conduct will institutionalize the gap.

4.5 *Recursive nostalgia and the culture wars of 1995*

The Victorians were themselves nostalgic. A. W. N. Pugin's *Contrasts* (1836), Thomas Carlyle's *Past and Present* (1843), the Young England movement, and Ruskin's medievalism measured the industrial present against an idealized Gothic past. Neo-Victorianism is therefore a nostalgia for a nostalgia, a second-order restoration whose object was already a restoration.

Its immediate context was the American culture war of the mid-1990s. Gertrude Himmelfarb's *The De-Moralization of Society* (1995), published the same year as the novel, contrasted Victorian 'virtues' with modern 'values' and urged their recovery. William J. Bennett's *The Book of Virtues* (1993), a bestselling anthology of moral stories for children, was a primer in the old sense: a book designed to form character through narrative. Richard Herrnstein and Charles Murray's *The Bell Curve* (1994) argued that intelligence is substantially heritable; that it increasingly sorts Americans into a cognitive elite and an underclass; that compensatory programmes such as Head Start had produced few durable cognitive gains; and that genes probably contribute to the average difference in measured IQ between Black and white Americans, a claim they presented as agnosticism about proportions. The American Psychological Association's task force on intelligence (Neisser et al. 1996) found no direct evidence for a genetic explanation of that difference. Whether or not in response to *The Bell Curve*, the novel's logic runs against it on two counts. It decouples the phyle from race: the Neo-Victorians admit members of any ancestry who accept their code, and the patron of the Primer, Alexander Chung-Sik Finkle-McGraw, carries mixed heritage in his very name. And its plot is a compensatory education that works: a thete child, formed by a book and a voice, ends as a sovereign. In the diamond age stratification is cultural and formative, and therefore in principle reversible.

Yet the novel's plot exceeds its conservative rhetoric. Nell's reversal requires inputs that no thete culture supplies and no self-reform could generate. The plot is Bourdieusian where the rhetoric is Himmelfarbian. It demonstrates that culture is infrastructure, distributed as unequally as any Feed line, and that its redistribution requires something like theft. I call the resulting politics *formationist*: neither neoconservative nor egalitarian, it locates both the cause of inequality and its remedy in the distribution of formative attention.

4.6 *The founder's paradox*

The Primer exists because Finkle-McGraw has noticed a paradox in his own phyle. Those who built the Neo-Victorian order were not raised within it. They were formed by the disorder they repudiated, and their achievement depended on the independence that disorder compelled. Their children, raised within the order, will be well-mannered and, he fears, docile. To reproduce its founders, the phyle must introduce into its children's education the very element it was founded to suppress. The Primer is commissioned to be subversive.

The commission arises from conversations that dramatize the paradox. Finkle-McGraw asks Hackworth whether his parents were subjects or whether he took the Oath. Hackworth took it at twenty-one, in Boston, and explains that as a child he knew two kinds of discipline, none at all and too much, and concluded that the previous century offered little worth emulating. The engineer who will build the Primer is himself a convert of the kind Finkle-McGraw fears his phyle can no longer produce. In the course of the commission the Duke tells Hackworth to ponder the meaning of the word subversive, and Hackworth realizes that the embodiment of the Victorian establishment is a subversive, unhappy that his own children are not. Much later Finkle-McGraw tells Carl Hollywood that he and his peers have devoted the past decade to the systematic encouragement of subversiveness, and that a young person who leaves for another phyle has merely outgrown credulity and will be ripe for New Atlantis once wise enough to judge it the best of tribes, a superlative cast in Pangloss's formula. The policy is a wager that exit breeds loyalty.

The paradox is Mill's. *On Liberty* (1859) warned against the 'despotism of custom', defended 'experiments of living', and observed that the amount of eccentricity in a society has generally been proportional to the genius and moral courage it contained. Mill's *Autobiography* (1873) supplied the case study. His father's experiment in total education produced a prodigy who, at twenty, found that analysis had worn away his capacity to feel, and who recovered through Wordsworth's poetry. The commission of the Primer is an attempt to compile Mill's individuality inside Arnold's culture. The rest of this dissertation argues that the novel's answer is that such individuality cannot be compiled. It can only be occasioned.

CHAPTER

V

The Confucian Mirror

Judge Fang, Dr. X, and the Celestial Kingdom

THE CELESTIAL KINGDOM is not the Neo-Victorians' antagonist but their reflection. Both are restorations that treat ritual propriety as a social technology for reproducing virtue, and the novel's moral architecture depends on their symmetry. This chapter reads the Confucian strand through its classical sources and its nineteenth- and twentieth-century contexts, and weighs the charge of techno-Orientalism.

5.1 *Two ritual orders*

The Confucian *li* (ritual propriety) and Victorian manners occupy the same structural position: codified conduct through which persons are cultivated and hierarchies made bearable. The *junzi*, the exemplary person of the *Analects*, and the Victorian gentleman are both ideals of self-command displayed in deportment. Filial piety and Victorian deference both locate the moral life in relations of obligation rather than in autonomous choice.

Max Weber's *The Religion of China* (1915) supplies the comparative frame. Weber contrasted Puritan rationalism, which sought the rational mastery of the world, with Confucian rationalism, which sought rational adjustment to it. The novel maps the contrast onto its two great restorations. New Atlantis is Baconian mastery, engineering its way to dominion; the Celestial Kingdom is Confucian adjustment, seeking harmony through ritual order and agrarian self-sufficiency. Both define themselves against the same enemy, the anomie of the Coastal Republic and the thete districts, a condition Émile Durkheim (1893; 1897) identified with the dissolution of the norms that regulate desire.

5.2 *Mencius, Xunzi, and the indigo*

The classical tradition divided over human nature in terms that bear directly on the Primer. Mencius held that human nature is good, possessed of four 'sprouts' of benevolence, righteousness, propriety, and wisdom that education need only nourish (*Mengzi* 2A6), and likened nature's tendency toward goodness to water's tendency to flow downward (6A2). Xunzi held that human nature is bad and that goodness is the product of deliberate effort: the warped board must be steamed and pressed against the straightening frame before it becomes straight (*Xunzi*, 'Xing e'). Rousseau is a Mencian. The Primer is a Xunzian instrument, an apparatus of artifice applied to unpromising material.

Xunzi's essay 'Encouraging Learning' ('Quanxue') opens with a figure that states the Primer's aspiration exactly: blue dye is extracted from the indigo plant yet is bluer than the plant; ice is made from water yet is colder than water. The pupil formed by learning surpasses the source of her formation. The Primer's ultimate success is to produce a pupil bluer than its indigo. Chapter 10 argues that what makes this possible is precisely what Xunzi's straightening frame cannot supply.

The novel stages this debate in Judge Fang's chambers. Deliberating over Harv, Miss Pao cites the *Analects* on rotten wood, which cannot be carved (5.10), and on the wise of the highest class and the stupid of the lowest, who alone cannot be changed (17.3). Judge Fang asks whether the boy is rotten wood, observing that his father certainly was, and Chang turns the court to the girl, who can still be saved. Verse 17.3 immediately follows the novel's epigraph, *Analects* 17.2, on nature and practice. The book's threshold asserts the power of formation; the next verse, spoken inside the book, asserts its limits. Coyote's Theorem (Chapter 10) can be read as a commentary on the pair.

5.3 *Judge Fang's jurisprudence*

Judge Fang, the Confucian magistrate before whom the thete cases of the Leased Territories come, descends from the scholar-magistrates of Chinese judicial fiction familiar in the West through Robert van Gulik's Judge Dee novels (1949–68), themselves built on the Tang official Di Renjie. His decisive ruling is to leave the stolen Primer with Nell rather than restore it to its owner, and the novel gives his reasoning in full. In deliberation he cites *Analects* 2.3: led by regulations and punishments, the people evade them and feel no shame; led by virtue and ritual, they acquire shame and order themselves. Miss Pao objects that the court has the power to punish but not to raise children. Fang names the dilemma of his office: the Coastal Republic built its judiciary on the Confucian model, yet such a system cannot function in a society that does not hold Confucian precepts, and the *Great Learning* makes the cultivation of the person the root of everything. He resolves it on two grounds, one formal and one canonical. The victim has not asked for his property back, and *Analects* 15.38 holds that in teaching there should be no distinction of classes. Harv, who stole the book, is sentenced to six strokes of the cane, five suspended because he gave it to his sister. The ruling subordinates property to formation; it is the novel's clearest statement of paideiarchy from the bench.

The same bench is not gentle, and the novel names what it does. Fang sentences Bud to death. He has Harv caned twice, the first time for beating the man suspected of molesting Nell. He sentences Hackworth to sixteen strokes and ten years' imprisonment, suspending all the strokes but one on condition that Hackworth give Dr. X the key to the Primer's data, so that copies can be made for the small children crowding the orphanages; the decade Hackworth then spends among the Drummers answers Fang's assurance that the sentence will be served one way or another. Interrogating PhyrePhox, a CryptNet member suspected of trafficking infant girls, Fang has Miss Pao deliver pain through nanosites lodged on the prisoner's auditory nerve. The caning was topical in 1994–95: that year Singapore caned Michael Fay, an eighteen-year-old American, for vandalism, provoking a transpacific debate about 'Asian values' that Lee Kuan Yew articulated in his interview with Fareed Zakaria ('Culture Is Destiny', 1994). Fang's later move from the Coastal Republic to the service of the Celestial Kingdom enacts the rectification of names (*Analects* 13.3), and the novel supplies his precedent: brooding on the Republic's corruption, he recalls the *Analects*' praise of Qu Boyu, who held office when his state was well governed and rolled up his principles when it was not (15.7). His defection is fidelity by other means. It is also the fidelity of a magistrate who tortures.

5.4 *Dr. X: the mandarin as strategist*

Dr. X, whose very name designates an unknown, is the novel's most effective strategist, and his two projects divide sovereignty into its material and formative halves. The Seed would give China material autarky. The Mouse Army, several hundred thousand abandoned girls raised on mass-compiled copies of the Primer, would give it a formed population devoted to its formation.

The first project answers Joseph Needham's question (*Science and Civilisation in China*, 1954–) as to why the scientific revolution occurred in Europe rather than China, and reverses the 'great divergence' that Kenneth Pomeranz (2000) traced to coal and colonies. It recalls the Maoist doctrine of self-reliance and Gandhi's *swadeshi*. The second project answers a demographic wound, and the historical wound it transposes had authors. Amartya Sen's 'More Than 100 Million Women Are Missing' (1990) estimated the worldwide shortfall of women relative to equal treatment and attributed it chiefly to the neglect of girls and women in nutrition and health care, with China among the largest cases. Under the one-child policy (1980–2015) the Chinese state's birth-planning apparatus enforced quotas through fines and coerced sterilizations and abortions; under that pressure, and with sons preferred, many families abandoned infant daughters. Kay Ann Johnson's *China's Hidden Children* (2016) documents those abandonments, together with cases in which family-planning officials seized children born outside the plan and placed them in orphanages. Sex-selective abortion, and in some places infanticide, widened the deficit. The novel's Mouse Army is gathered from girls its own society discarded, and Dr. X converts that wound into a cohort: a would-be state acquires its most loyal population by raising the children others cast off. That is paideiarchy in its starkest form. In Platonic terms the Mouse Army is a guardian class reared in common (*Republic* 457c–461e); in Confucian terms it is a population ordered by ritual instruction.

The novel supplies its own genealogy for the cohort. Judge Fang's case against PhyrePhox begins with a riverboat overloaded with peasants fleeing famine and strife in the interior, several of them women carrying female infants; constables then find twenty-four Han infant girls in a supposedly vacant flat in the Leased Territories, tended by peasant women recruited by a Han gentleman who is never identified. Fang explains his interest by invoking the Boxer-era rumour that missionary orphanages were places where European doctors took the eyes of Chinese infants to make medicine, a rumour that helped incite the attacks on missions and converts in 1900 (Section 5.5). The cohort's books come from his sentence on Hackworth: the key to the Primer's data, so that copies can be made, in the hundreds of thousands, for the small children crowding the orphanages. Hackworth objects at once that the Primer is in large part a ractive requiring adult performers, and that more than a few copies would overwhelm the arrangements for paying them. The device's designer warns the court that the remainder does not scale. Corollary 1 of Chapter 10 is stated first by Hackworth.

Dr. X's single failure is instructive. He can replicate the Primer's code without limit. He cannot replicate Miranda. His girls are formed but not individuated, devoted but leaderless, and the sovereign they require must come from outside them.

Dr. X states the Celestial theory of the Seed to Hackworth in terms that confirm the paideiarchy thesis from the other side. The Outer Tribes, he argues, lack order within their own minds and must impose it from above, through control of the Feed, withholding the Seed for fear of the weapons, viruses, and drugs their people would make with it. In the Celestial Kingdom order begins in the disciplined mind and extends to family, village, and state, so that in its hands the Seed would be harmless. This is the *Great Learning*'s sequence of self-cultivation, fam-

ily, and state turned into a theory of technology, and it makes decentralized power conditional on prior formation. Dr. X also roots the Seed in agrarian tradition: China has lived by the seed for five thousand years, and the Confucian hierarchy, as he puts it, ranked the peasants who planted it highest, a point he supports with a maxim from the *Great Learning* that producers should be many and consumers few. The Seed is not only the anti-opium; it is *ti*.

5.5 *Boxers redux*

The historical Yihetuan combined martial ritual, spirit possession, and a belief in invulnerability with violence against what they took to be foreign: railways, telegraphs, missionaries, and above all Chinese converts to Christianity (Esherick 1987). In 1900, chiefly in Zhili and Shanxi, Boxers killed tens of thousands of Chinese Christians and some two hundred foreign missionaries and members of their families; at Taiyuan the Shanxi governor Yuxian had foreign missionaries and their families put to death. In June the Qing court declared war on the foreign powers, and Qing troops joined the Boxers in besieging the Beijing legations until August. The Eight-Nation Alliance that relieved the legations looted the capital, and its punitive expeditions killed Chinese civilians along with suspected Boxers; in July Russian troops drove thousands of Chinese residents of Blagoveshchensk into the Amur, where they drowned. The Boxer Protocol of 1901 imposed an indemnity of 450 million taels. Paul Cohen's *History in Three Keys* (1997) distinguished the Boxers as event, as lived experience, and as myth subsequently appropriated by Chinese nationalisms of every stripe. *The Diamond Age* belongs to the third key. Its Fists are the myth reactivated, a ritual collectivity whose effervescence aboveground mirrors the Drummers' computational collectivity below. The history fixes what such effervescence did: it was turned first on a domestic minority marked as foreign by its faith, and it was answered by foreign reprisals against civilians. Both collectivities are figures of the crowd that the Primer's individualizing pedagogy exists to resist.

5.6 *Techno-Orientalism and its limits*

Edward Said's *Orientalism* (1978) exposed the Western construction of the East as an object of knowledge and power. David Morley and Kevin Robins (1995) identified a 'techno-Orientalism' that projects Western anxieties about technological futurity onto East Asia, and the collection edited by David S. Roh, Betsy Huang, and Greta A. Niu (2015) traced the trope through speculative fiction. The novel is not innocent of it. Dr. X's inscrutability, the frenzy of the Fists, and the labyrinthine opulence of its Shanghai belong to a familiar repertoire.

Two considerations complicate the charge. First, the novel grants its most morally serious reasoning to a Confucian magistrate and its strategic victory to the Celestial Kingdom; the Neo-Victorians emerge as the more costumed and the less clear-sighted party. Second, its exoticism is symmetrical. It assigns to China its Confucian and Boxer past and to the West its Victorian past, and locates both pasts in the future. The novel's deeper Orientalism is chronological: it renders every civilization as a revival. That device is the strongest evidence for the paideiarchy thesis. In a

market of civilizations, each competes by reissuing the formative repertoire of its most self-confident century.

Nor is the Celestial interior idealized. Travelling there, Hackworth sees girls carried in sedan chairs, absorbed in their Primers and studying nanotechnical diagrams; one has slipped off her slipper, and her foot is several inches too short, evidently bound. The restoration that teaches its daughters the secrets of atoms has, it seems, revived the binding of their feet. Read beside Fang's torture of PhyrePhox, the detail shows that the novel's symmetry between its two restorations extends to their cruelties.

CHAPTER

VI

A Propædeutic Enchiridion

The Primer and the Philosophy of Education

THE PRIMER CONDENSES the history of Western pedagogy into a single object, and its title is a genealogy in miniature. This chapter reads the device against its originators, from Plato's mythcurriculum to Papert's microworlds, and identifies the paradox, first stated by Kant, that its commission is designed to solve.

6.1 *The title as genealogy*

Each term of the device's full title names a pedagogical form. A *primer* was originally a lay devotional book and became, in *The New England Primer* (c. 1690), the instrument through which literacy and Calvinist doctrine were taught as one. *Illustrated* recalls the Victorian revolution in the picture book. *A Young Lady's* invokes the conduct literature through which girls were formed for domesticity, of which Hannah More's *Strictures on the Modern System of Female Education* (1799) is representative. *Propædeutic* denotes preparatory instruction, that which precedes *paideia* proper.

Enchiridion is the richest term. It names the *Enchiridion* of Epictetus, compiled by Arrian, the Stoic handbook whose first teaching is the discrimination between what is up to us and what is not. It also names Erasmus's *Enchiridion militis Christiani* (1503), which exploited the word's double sense: a thing held in the hand, hence both a handbook and a dagger. The Primer is both. It teaches Nell Stoic self-command and, quite literally, how to fight.

6.2 *Plato's wager*

Plato's *Republic* (377a–c) holds that the stories told to children shape their souls more powerfully than any later instruction, and that a city must therefore supervise its myths. The noble lie (414b–415d) is a founding myth designed to secure civic loyalty. The Primer realizes Plato's program technologically: it is a myth-engine that composes, in real time, the story a particular soul requires. Its originality lies in its detachment. It serves no city. It is a Platonic instrument loose in the world, and whoever possesses it holds the most potent formative power Plato could imagine.

The *Meno* (82b–85b) supplies the complementary scene. Socrates leads an enslaved boy, by questions alone, to a geometrical truth, and concludes that the capacity for knowledge belongs to every soul regardless of station. Nell is Meno's slave-boy raised to a principle. The Primer's interrogative method is Socratic, and the demonstration it performs is the dialogue's: the lowest station in the social order conceals no deficit of capacity.

6.3 *Rousseau's Sophie, Wollstonecraft's rejoinder, Kant's paradox, Arendt's crisis*

Rousseau's *Émile* (1762) imagined a tutor who orchestrates his pupil's environment so that the pupil learns from things rather than words, and proposed a 'negative' education that shields nature from premature corruption. Its fifth book educated Sophie, Émile's intended wife, to please men and to defer to their judgment. Mary Wollstonecraft's *A Vindication of the Rights of Woman* (1792) answered that women, as rational beings, must be educated for independence. The Primer, commissioned by a patriarch for a 'young lady' with the express intention of making her subversive, is a Wollstonecraftian correction lodged inside a conservative project, and its effects escape its commissioner's intentions in each of its three original recipients.

Kant's *Lectures on Pedagogy* (1803) posed the problem the commission exists to solve: how to cultivate freedom under constraint, how to unite submission to lawful compulsion with the capacity to use one's own freedom. His essay 'What Is Enlightenment?' (1784) defined enlightenment as emergence from self-incurred immaturity and indicted the guardians who keep their charges in tutelage. A Primer is a guardian designed to abolish its own necessity.

Hannah Arendt's 'The Crisis in Education' (1958) restated the paradox in its most useful form. Education, Arendt argued, must be conservative precisely in order to protect what is new and revolutionary in each child, preparing the young in advance for the task of renewing a common world. She grounded the claim in *natality*, the fact that each birth introduces a being capable of beginning something unforeseen (*The Human Condition*, 1958). Finkle-McGraw's commission has exactly this structure: a conservative instrument whose purpose is to protect the capacity to begin. Hegel's account of formative labour in the *Phenomenology of Spirit* (1807), in which the bondsman attains self-consciousness through work while the master stagnates in enjoyment, adds a further turn: the formative advantage lies with the dispossessed.

6.4 *From Pestalozzi to the Dynabook*

The Primer's techniques gather two centuries of progressive pedagogy. Pestalozzi's method (1801) educated head, heart, and hand together. Froebel's 'gifts' (1826) made objects of play into instruments of instruction; the Primer's toy companions are Froebelian gifts with voices. Montessori's prepared environment (1909) arranged materials so that the child teaches herself. Dewey's *Democracy and Education* (1916) made experience the medium of growth. Vygotsky's zone of proximal development (*Mind in Society*, 1978), and the 'scaffolding' described by Wood, Bruner, and Ross (1976), name the Primer's core competence: it calibrates each challenge to what the child can achieve with assistance and withdraws assistance as competence grows. Jerome Bruner's distinction between the paradigmatic and narrative modes of thought (1986) names its method: it teaches logic through story.

Seymour Papert's *Mindstorms* (1980) drew the distinction on which the Primer's arc turns: whether the computer programs the child or the child programs the computer. The Primer begins by programming Nell, shaping her through stories she did not choose, and ends by teaching her to program, most explicitly in the Castle Turing sequence. Alan Kay's Dynabook (1972), 'a personal computer for children of all ages', is the Primer's direct engineering ancestor. Benjamin Bloom's '2 Sigma Problem' (1984) reported that students tutored one-to-one performed about two standard deviations above conventionally taught peers, and challenged researchers to find group methods as effective as tutoring. The Primer is a solution to Bloom's problem, and its dependence on human ractors registers Stephenson's intuition, consistent with Bloom's own data, that the two sigmas reside in the tutor.

6.5 *Fairy-tale formation*

Vladimir Propp's *Morphology of the Folktale* (1928) reduced the wondertale to a sequence of functions; Joseph Campbell's *The Hero with a Thousand Faces* (1949) generalized the quest into a monomyth. Bruno Bettelheim's *The Uses of Enchantment* (1976) argued that fairy tales help children externalize and master inner conflict, and the Primer operates on Bettelheim's principle. It allegorizes Nell's domestic terror as the Dark Castle, her abusers as its tyrants, and her guardians as four toy companions (Dinosaur, Duck, Peter, and Purple) whose temperaments partition the virtues it means to cultivate: martial spirit, care, cunning, and arcane knowledge, roughly Plato's *thumos*, a nurturing *eros*, the Greek *mētis* anatomized by Detienne and Vernant (1974), and *sophia*. Jack Zipes's *Fairy Tales and the Art of Subversion* (1983) analyzed the tale as an instrument both of socialization and of resistance; a subversive Primer is Zipes's thesis built in hardware. Tolkien's 'On Fairy-Stories' (1947) named the sudden joyous turn of a tale's resolution 'eucatastrophe'; the novel's ending is one.

The Victorian intertexts are dense. George MacDonald's *The Princess and the Goblin* (1872) gives Princess Irene a great-great-grandmother who guides her by an invisible thread; the Primer's network tie to Miranda is MacDonald's thread spun in fibre. Charles Kingsley's *The Water-Babies* (1863) conducts a chimney sweep's moral education underwater; the Drummers invert it.

Lewis Carroll's Alice inhabits an allegorical world of logic games devised by a mathematician whose *Symbolic Logic* (1896) was itself a primer. Dickens's Little Nell of *The Old Curiosity Shop* (1841), the sentimental child whose death famously moved a nation and famously amused Wilde, is the unhappy namesake. Stephenson's Nell does not die. She fights, studies, and rules. The name is a correction.

6.6 *Castle Turing and Lady Lovelace's objection*

Alan Turing's 'Computing Machinery and Intelligence' (1950) contains two proposals that the Primer engages. The first is the 'child-machine': rather than simulate an adult mind, Turing suggested, build a machine like a child's and educate it. The Primer inverts the proposal; a machine educates a child. The second is Turing's reply to 'Lady Lovelace's objection'. In her notes on the Analytical Engine (1843), Ada Lovelace wrote that the Engine has no pretensions to originate anything and can do only what we know how to order it to perform. Turing contested the claim; the Primer concedes it. Whatever originates in the Primer's dealings with Nell enters through its human performer or through Nell herself.

In the Castle Turing sequence, Princess Nell learns that the operations of an apparently intelligent castle reduce to the mechanical manipulation of symbols, a narrative staging of Turing's analysis of computation (1936). She learns that intelligence can be mechanism. The novel teaches her, more slowly, that the voice that taught her this is not. Lovelace herself is the Neo-Victorian girl's historical archetype: Byron's daughter, schooled in mathematics by a mother determined to suppress the poetic temperament, who arrived at what she called a 'poetical science'. William Gibson and Bruce Sterling's *The Difference Engine* (1990), the steampunk novel of a Victorian computational revolution, is *The Diamond Age*'s sibling.

The Primer turns the inquiry on itself. Studying Castle Turing, Nell works out that its soldiers are wind-up men and its Duke probably one too, and she sets herself two problems: how the lock on a door works, and whether the correspondent sending her messages is human or mechanical. That is Turing's test, posed by a child to a book. The romance answers it at its summit. Princess Nell, now Duchess of Turing, has studied the Cipherers' Market, where messages are processed by consulting rules in books, and she rides to the castle of King Coyote to obtain the twelfth key and to learn whether his great thinking machine, Wizard 0.2, is a Turing machine. It is, he tells her, the most powerful ever built, and the Land Beyond grew from seeds he invented. But the messages that govern the land he reads and answers himself, so the Land Beyond is not really a Turing machine at all. It is a person, a few people in fact, and now it is hers. He gives her his library, the rules for programming the machine and the books on making atoms build themselves into worlds, and charges her to make new worlds for others to explore and conquer. The Primer's last lesson is its own Mechanical Turk structure, disclosed to the pupil at the moment she inherits the means of making worlds.

6.7 *Books that read their readers*

Michael Ende's *The Neverending Story* (1979) imagined a book that contains its reader and threatens to consume him. Borges's 'The Garden of Forking Paths' (1941) imagined a novel branching through every possibility; 'Tlön, Uqbar, Orbis Tertius' (1940) imagined a fictional world whose artifacts invade and replace the real one. The Primer is a Tlönian object. Its romance of Princess Nell eventually acquires an army in the world, and its fictional sovereignty becomes an actual polity. The *Thousand and One Nights* supply the oldest frame. Scheherazade tells stories nightly to preserve a life; Miranda, voicing the Primer through the nights of a decade, does the same.

The Tlönian reading is literal. Late in the Primer's romance Princess Nell crosses the trail of a Mouse Army, mice riding cats as knights ride horses. In King Coyote's library she finds the floor carpeted with them, and they spell out that they are enchanted and ask her help. Of all the creatures of the Land Beyond, only the mice are not figments of King Coyote's machine; they exist independently and have long sought their Queen. When Princess Nell reads the spell of disenchantment, they become a host of some hundreds of thousands of girls, a few years younger than she, who swear fealty to Queen Nell; she then sets thousands of them to copying her books in a scriptorium, a figure for the replication of the Primer itself. In the novel's last movement, the real army of girls raised on Dr. X's copies swears allegiance to Nell, calling itself, for reasons Carl Hollywood cannot fathom, the Mouse Army.

The most exact precedent is Goethe's *Wilhelm Meisters Lehrjahre* (1795–96). Wilhelm's apparently aimless apprenticeship proves to have been observed and steered throughout by the Society of the Tower, which finally presents him with his certificate of apprenticeship. The Primer is the Tower miniaturized and given a voice.

6.8 *Conditioners and liberators*

The Primer's power cuts both ways, and the tradition anticipated both edges. C. S. Lewis's *The Abolition of Man* (1943) warned of a generation of 'Conditioners' who would manufacture the values of posterity, producing 'men without chests': persons lacking the trained sentiments that mediate between intellect and appetite. Huxley's *Brave New World* (1932) imagined moral formulas implanted through sleep-teaching; Skinner's *Walden Two* (1948) offered the behavioural engineering of children as benevolent policy. Paulo Freire's *Pedagogy of the Oppressed* (1968) opposed a 'banking' model, in which knowledge is deposited in passive recipients, to a problemposing education that awakens critical consciousness; Ivan Illich's *Deschooling Society* (1971) imagined learning webs in place of institutions.

The Primer holds all of these potentialities in one object. With Miranda in it and the world around it, it is Freirean. Mass-compiled for Dr. X's girls, it is Lewis's Conditioner. The difference does not lie in its code.

CHAPTER

VII

The Ractor in the Machine

Voice, Care, and the Uncompilable Remainder

THE PRIMER IS not an artificial intelligence. It is a Mechanical Turk, and the person inside it is the novel's moral centre. This chapter analyzes the ractor as a figure of hidden labour, of commodified care, and of a surplus no compiler can produce.

7.1 *A Turk in the book*

Wolfgang von Kempelen's chess-playing automaton (1770) astonished the courts of Europe for decades; a human player was concealed in its cabinet. Edgar Allan Poe's 'Maelzel's Chess-Player' (1836) reasoned that the machine could not be a pure mechanism precisely because chess is not determinate: a machine runs a fixed course, whereas every move of a game depends on the unforeseen move before it. Amazon named its crowdsourced labour market Mechanical Turk (2005) and marketed it as 'artificial artificial intelligence'. Mary L. Gray and Siddharth Suri's *Ghost Work* (2019) and Kate Crawford's *Atlas of AI* (2021) documented the concealed human labour on which automated systems depend.

The Primer's architecture is the Turk's. Its synthetic intelligence manages narrative logic, perception of the child's surroundings, and adaptation, but for the full-bodied voicing of its characters it subcontracts to anonymous ractors who perform in real time through the mediatronic network, paid for their time and ignorant of whom they address. John Searle's Chinese Room (1980), a system that manipulates symbols without understanding them, is here inverted: the Primer is a room that does not understand containing a person who does, and understanding enters the system through her. The Turing test is inverted too. The Primer passes for a mother because it partly is a person.

Hackworth's own account of the device fixes the division of labour. On first opening, the Primer bonds with the girl who holds it, imprinting her face and voice, and thereafter interprets every event in relation to her, maintaining a psychological map onto which it projects universal ideas drawn from folklore. That is personalization by compilation: a specification fixed in advance, indexed to one child. What the map cannot supply, the living voice that speaks its charac-

ters, the design leaves to paid human performers, and Hackworth's warning to Judge Fang about mass copies (Section 5.4) shows that he knew it.

The architecture anticipates the present. Large language models are aligned through reinforcement learning from human feedback (Christiano et al. 2017; Ouyang et al. 2022), in which the judgments of many human raters are distilled into a model's dispositions. Contemporary tutoring systems are diffused Mechanical Turks: their voice is an aggregate of countless unseen evaluators, none of whom addresses any particular child. Stephenson's Primer concentrated the hidden human into one person for one child. Its successors disperse her across a workforce. Chapter 10 argues that the difference is decisive.

7.2 *Miranda: The Tempest inverted*

Miranda's name is Shakespeare's. In *The Tempest* (c. 1610–11) Miranda is the pupil, educated on an island by a magus whose power resides in his books; her cry at the sight of strangers, 'O brave new world' (5.1), gave Huxley his title. Caliban, taught language by Prospero and Miranda, complains that his profit from the lesson is that he knows how to curse (1.2). Octave Mannoni's *Psychologie de la colonisation* (1950) and Aimé Césaire's *Une Tempête* (1969) made the play a parable of colonial pedagogy, in which instruction is the instrument of subjection.

Stephenson inverts every term. His Miranda is the teacher, concealed inside the magic book rather than educated by it. Her pupil is a Caliban, a child of the propertyless margins, who acquires language not to curse but to found. And where Prospero, abjuring his art, vows to drown his book 'deeper than did ever plummet sound' (5.1), Nell descends beneath the sea to recover the voice behind hers. The Latin gerundive *miranda* means 'she who is to be wondered at'. The novel locates wonder not in the device but in the person within it.

7.3 *Emotional labour and the commodification of mothering*

Miranda's work is emotional labour in Hochschild's sense (Chapter 2), and of the most intimate kind: the paid production of a mother's regard. Federici's account of social reproduction explains its economic invisibility; Fraser's crisis of care explains its precarity. Charlotte Perkins Gilman's *Women and Economics* (1898) and *Herland* (1915) imagined the professionalization and collectivization of motherhood, a society in which child-rearing is the supreme civic vocation, entrusted to specialists. Shulamith Firestone's *The Dialectic of Sex* (1970) demanded technologies that would free women from the tyranny of reproduction. The Primer partly realizes both programs. It professionalizes the instructional half of mothering and automates much of it. It fails, instructively, to automate the rest.

What forms Nell is not Miranda's contracted labour but the increment beyond it. Over a decade Miranda gives the Primer more hours, more invention, and more of herself than her position requires, organizes her life around a child whose face she has never seen, and finally sets out to find her. The structure inverts Marx's analysis of surplus value in *Capital* (1867). There, labour yields more than its wage and capital appropriates the excess. Here, care exceeds its wage

and the excess is appropriated by no one; it passes to the child as a gift. Marcel Mauss's *The Gift* (1925) analyzed the triple obligation to give, to receive, and to reciprocate, and the novel completes Mauss's circuit when Nell, having received the gift, returns it by descending to rescue the giver.

The novel specifies the conditions of this labour. Miranda works at the Theatre Parnasse, performing in one of twelve egg-shaped 'body stages' that generate three-quarters of the theatre's cash flow, her movements captured through a grid of nanosite tattoos that she checks for failures before each shift. She arrives early, takes overtime when a well-heeled client wants a ractive stretched into the night, and hopes to attract the attention of London's sophisticated paying audience. A month or two into Nell's account, she signs off from a Princess Nell session and realizes that she has been performing for eight hours without a break. No contract required it.

7.4 *Aura, storyteller, polyphony*

Benjamin's essay on reproducibility supplies the clearest account of what Dr. X cannot copy. The Primer's code is perfectly reproducible, and Dr. X reproduces it by the hundred thousand. Miranda's performance has what Benjamin called aura, a unique existence in time and place, and it does not survive copying. Benjamin's 'The Storyteller' (1936) lamented the decline of the art of passing experience from mouth to ear, and of the storyteller as one who counsels; the Primer revives the storyteller by network, in what Walter Ong (1982) called secondary orality, a spoken intimacy sustained by literate technology. Brenda Laurel's *Computers as Theatre* (1991), which proposed dramatic theory as the foundation of interactive design, is the ractive industry's theory avant la lettre; Janet Murray's *Hamlet on the Holodeck* (1997), with its triad of immersion, agency, and transformation, describes what Nell undergoes.

Mikhail Bakhtin's account of polyphony (1929; rev. 1963), the novel as a plurality of independent voices, applies with a twist. Miranda voices every character in Nell's Primer. It is a polyphonic novel whose many voices issue from a single throat, and its unity is the unity of one performer's regard.

7.5 *Wire mothers and cloth mothers*

Harry Harlow and his colleagues at the University of Wisconsin separated infant rhesus macaques from their mothers within hours of birth and offered them two surrogates, one of bare wire that dispensed milk and one of soft cloth that dispensed nothing ('The Nature of Love', 1958). The infants fed at the wire and clung to the cloth. In later studies the laboratory reared some infants in total social isolation for months, which left them with lasting, severe disturbance (Harlow and Harlow 1962); and even the cloth-reared monkeys grew into socially and sexually abnormal adults, several of whom, as mothers, neglected or attacked their own infants. Contact with peers partly repaired the damage. René Spitz's study of 'hospitalism' (1945) found that infants in a foundling home who were fed and kept clean but had no individual caregiver declined in development and died at a markedly higher rate. John Bowlby's *Attachment* (1969) theorized the bond. The children studied by the Bucharest Early Intervention Project (Nelson, Fox, and Zeanah 2014) lived in Romanian state institutions whose population had swollen after Nicolae Ceaușescu's regime banned abortion and most contraception in 1966 (Decree 770) and pressed families to bear children many could not support; the project documented the developmental costs of that institutional rearing and their partial reversal through placement in foster families.

The Diamond Age stages Harlow's experiment at civilizational scale, and the full record sharpens the analogy. The Feed is the wire mother of the whole society: abundant nourishment, no contact. The Mouse Army's Primers are wire mothers of formation: abundant instruction, delivered at scale, without individuated regard. Nell's Primer, however, is not simply a cloth mother, for Harlow's later findings show that comfort without a responsive other also fails. It is a cloth mother with a person inside it, a social partner who answers, which is why Section 7.6 defines the remainder as person-indexed and reciprocally binding rather than as comfort. The outcomes differ as the full record predicts. The Mouse Army is competent, disciplined, and devoted, but its devotion seeks an object outside itself. Nell, uniquely, becomes such an object.

7.6 *The culture of the feelings*

Mill's crisis of 1826, recounted in the fifth chapter of his *Autobiography*, is the Victorian archetype of compiled education: a curriculum of unexampled rigour produced an intellect without feeling, and the habit of analysis, Mill concluded, tends to wear the feelings away. Carlyle's 'Signs of the Times' (1829) had diagnosed an Age of Machinery in inward as well as outward life, and opposed the 'mechanical' to the 'dynamical', the spontaneous sources of conviction no machinery supplies. Lewis's 'men without chests' restates the diagnosis. The Primer without Miranda is James Mill's curriculum. Miranda is Wordsworth.

The *uncompilable remainder* can now be defined. It is that component of formative address which is (i) *time-indexed*, given in the moment and incapable of precomputation; (ii) *person-indexed*, addressed to this child and shaped by her responses; (iii) *non-contractual in its decisive increment*, exceeding what any arrangement requires; and (iv) *reciprocally binding*, creating an obligation that becomes a bond. Each property defeats compilation, because a compiler produces

outputs from a specification fixed in advance, and the remainder consists in exceeding specification. A scripted surprise is a routine; a contracted devotion is a service. The remainder can be occasioned, protected, and rewarded. It cannot be manufactured.

7.7 *Machines raised, and children raised by machines*

Ted Chiang's *The Lifecycle of Software Objects* (2010) inverts the Primer's premise and confirms its conclusion. Its human keepers raise digital creatures over years and discover that there is no shortcut: a mind worth having requires the same sustained, embodied, uneconomic attention as a child. Kazuo Ishiguro's *Klara and the Sun* (2021) poses the question the Primer suppresses: what if the machine itself can love? Klara's devotion to her sick child is that novel's moral centre, and Ishiguro leaves open whether it is love or its flawless simulation. Donna Haraway's 'A Cyborg Manifesto' (1985) observed that cyborgs are the illegitimate offspring of militarism and patriarchal capitalism, and that illegitimate offspring are often exceedingly unfaithful to their origins. The stolen Primer is such an offspring: commissioned by an aristocrat to perpetuate his order, it raises a thete who founds another. N. Katherine Hayles's title *My Mother Was a Computer* (2005) could serve as an epitaph for Nell's childhood, with one essential qualification: her mother was a computer with a woman inside it.

CHAPTER

VIII

Drummers, Seed, and the Accursed Share

Collective Mind and Sacrifice

Beneath the dignified surface of the diamond age lies a collectivity that is the Primer's antithesis: a choir without a voice. This chapter reads the Drummers through Bataille's general economy and the mythology of the Grail, places them in the history of the group mind, and interprets the climax as an inverted descent.

8.1 *Computation in the flesh*

The Drummers are a subaqueous collective whose members, linked by nanosites exchanged through bodily fluids, compute in concert amid drumming, trance, and erotic communion. Their distributed cognition is harnessed to the design of the Seed; Hackworth spends a decade submerged within it, and Miranda is later drawn into it. The fiction extrapolates from science contemporaneous with the novel. Leonard Adleman's demonstration of DNA computing (1994) solved a combinatorial problem with molecules in solution. Rudy Rucker's *Wetware* (1988) imagined minds propagated through organic substrates. Stephenson's own *Snow Crash* (1992) imagined a metavirus spread through blood and language that reduces its hosts to a glossolalic hive; the Drummers are its sequel. Douglas Hofstadter's 'Ant Fugue' in *Gödel, Escher, Bach* (1979) introduced Aunt Hillary, an ant colony that converses as a mind though no ant understands anything, and Marvin Minsky's *The Society of Mind* (1986) modelled intelligence as the interaction of mindless agents. The Drummers are Aunt Hillary made of people.

The Drummers and the Primer are exact opposites. The Primer individualizes through a single voice addressed to a single child. The Drummers dissolve individuals into a computation that addresses no one.

8.2 *The accursed share*

Georges Bataille's *The Accursed Share* (1949) proposed a 'general economy' in which every system produces more energy than growth requires and must expend the surplus unproductively: in sacrifice, festival, luxury, war, or eroticism. *Erotism* (1957) analyzed eroticism and death as the means by which discontinuous beings glimpse the continuity from which individuation separates them. In a post-scarcity order the surplus is general, and Bataille predicts it will be spent. The Drummers spend it. Their erotic fusion abolishes the discontinuity of persons, and their rite, which opens Part the Second, gathers the collective around a single woman at its centre, with men drawn into the ring around her, some willingly and some pushed or prodded by celebrants; at the climax that central place is Miranda's, her skin alight with nanosites until Nell's blood extinguishes them. The diamond age's accursed share is expended not in war, as Bataille feared, but in cognition. The Drummers are a potlatch of minds.

Durkheim's account of collective effervescence in *The Elementary Forms of Religious Life* (1912), Freud's pairing of Eros and the death drive in *Beyond the Pleasure Principle* (1920), Gustave Le Bon's *The Crowd* (1895), and Elias Canetti's *Crowds and Power* (1960), with its 'crowd crystals' around which masses precipitate, supply the remaining coordinates. The Fists aboveground and the Drummers below are two effervescent collectives, one political and one computational, and both are what the Primer's pedagogy of the singular voice exists to resist.

8.3 *Percival in the waste land*

The engineer who disappears into the depths is named John Percival Hackworth. In Chrétien de Troyes's *Perceval* (c. 1180s) and Wolfram von Eschenbach's *Parzival* (c. 1200–10), the innocent knight reaches the Grail Castle, fails to ask the question that would heal its wounded king, and wanders for years before returning to complete the quest. Hackworth's decade among the Drummers is a Grail wandering, and the Seed is his Grail: the instrument that would regenerate a waste land made barren by dependence. Jessie Weston's *From Ritual to Romance* (1920) read the Grail legend as the vestige of a fertility ritual; James Frazer's *The Golden Bough* (1890) had gathered the myths of the dying and reviving god and the sacrificed king; T. S. Eliot's *The Waste Land* (1922) built its desolation on both. The Drummers' rite, orgiastic, sacrificial, and devoted to the production of a seed, is Westonian vegetation ritual in nanotechnological form.

The novel adds a characteristically Stephensonian twist. Dr. X's charge to Hackworth arrives as the message of a fortune: seek the Alchemist. A decade later Colonel Napier, who has only lately learned of the Alchemist, is startled to hear that Dr. X asked it of Hackworth so long ago. It is Nell who solves the riddle at the end. The Drummers' collective is a wet net, usable for distributed computation, and Hackworth has been running a vast program of his own on it: he is the Alchemist, designing the Seed. Percival's unasked question receives an unexpected answer. The knight was the object of his own quest.

8.4 *Morlocks and the command line*

H. G. Wells's *The Time Machine* (1895) imagined humanity bifurcated into decorative Eloi above and labouring Morlocks below. Stephenson revived the figure in *In the Beginning... Was the Command Line* (1999) to distinguish those who consume technological surfaces from those who understand the machinery beneath. The Drummers are Morlocks of cognition: a submerged population whose labour, unseen by the dignified surface, produces the technology on which that surface's future depends. They are the diamond age's ghost work. Kingsley's water-babies were morally educated underwater; the Drummers are morally dissolved there.

8.5 *The group mind and its valuation*

Science fiction has more often exalted the group mind than feared it. Olaf Stapledon's *Last and First Men* (1930) and *Star Maker* (1937) imagined communal minds as stages of cosmic ascent; Theodore Sturgeon's *More Than Human* (1953) made a gestalt of outcasts the next step in evolution; Arthur C. Clarke's *Childhood's End* (1953) dissolved humanity's last generation of children into the Overmind. Pierre Teilhard de Chardin's *The Phenomenon of Man* (1955) gave the trajectory a theology, the noosphere converging on the Omega Point; Kevin Kelly's *Out of Control* (1994) gave it a cybernetics, the hive mind as a model for networked economies. Vernor Vinge's *A Fire Upon the Deep* (1992) explored pack minds with ambivalence, and his essay on the technological singularity (1993) located the posthuman future in the merger of minds and machines.

The Diamond Age dissents. Its group mind is regression rather than transcendence, a collective that computes brilliantly and speaks with no one's voice. Clarke's children are taken up into the Overmind; Stephenson's daughter descends into the collective to bring one person out.

8.6 *CryptNet: Masonry for hackers*

CryptNet, the secretive synthetic phyle whose designs on the Seed intersect the Drummers' computations, combines the graded secrecy of Freemasonry with the program of the cypherpunks. Its ancestors are the masonic lodges and Adam Weishaupt's Bavarian Illuminati (1776); Lessing's dialogues *Ernst und Falk* (1778–80), which imagined Masonry as an invisible society working behind visible ones; and the manifestos of Timothy May ('The Crypto Anarchist Manifesto', 1988) and Eric Hughes ('A Cypherpunk's Manifesto', 1993). Elizabeth Finkle-McGraw's flight into CryptNet is the commission's ironic fulfilment. Her grandfather wanted a subversive and received one, but her subversion takes the form of exit, in Hirschman's sense, rather than foundation, in Arendt's. She leaves an order. She does not begin one.

8.7 *The descent inverted*

The climax is a descent narrative, and its inversions are pointed. Orpheus descends for Eurydice and loses her by looking back; Nell descends for Miranda and brings her out. Prospero drowns his book; Nell enters the deep to retrieve the voice behind hers. The most resonant inversion is Eleusinian. In the *Homeric Hymn to Demeter*, the goddess of grain searches for her daughter Persephone, carried off to the underworld, and her grief withholds the seed from the earth until the daughter's return, partial and annual, restores fertility. The myth underwrote the mysteries of Eleusis and the agrarian cycle of descent and return. Stephenson inverts its direction of rescue: the daughter descends for the mother, and the recovery belongs to the same denouement in which the Seed enters the world. The reading is speculative, but it is exact. The novel ends on an Eleusinian mystery in which grain and voice return together.

The final pages bear the reading out. Nell fights her way into the Drummers' inner circle, where Miranda lies on the central altar with arms outstretched as if crucified, her skin alight with nanosites. Nell kisses her and bites through both their lips so that their blood mingles, and hunter-killer nanosites carried in Nell's blood destroy those in Miranda's: the gift returns as an exchange of blood, Mauss's reciprocity made literal. Carl Hollywood carries Miranda while Nell, sword drawn, leads them up through the tunnels and severs the passage behind them. As they swim for the surface, girls of the Mouse Army dive to bear them up into the light beneath New Chusan, and the cathedral bells ring on the mountain. The novel opened with bells ringing on the same mountain as Bud skated off to upgrade his skull gun. It closes on them as Nell rises with the woman who raised her.

CHAPTER

IX

A Comparative Constellation

Comparison here proceeds by problem rather than by influence. Each grouping below collects works that answer one of the novel's questions differently, and each isolates what is distinctive in Stephenson's answer.

9.1 *Fictions of formation*

Huxley's *Brave New World* (1932) is the Primer's negative: a formation engineered for stability, in which the World State's hatcheries chemically stunt lower-caste embryos, its conditioners train Delta infants with alarms and electric shocks to fear books and flowers, and hypnopaedia replaces the voice. Huxley's late *Island* (1962) is its positive, a humane pedagogy of attention on the island of Pala, overrun at the end by an oil-hungry neighbour. Skinner's *Walden Two* (1948) offers behavioural engineering as benevolence. Orson Scott Card's *Ender's Game* (1985) is the closest analogue in the genre's mainstream. Its Fantasy Game is adaptive narrative software that probes a gifted child's psyche on behalf of the state, and both novels culminate in a child commanding an army. The resemblance throws the difference into relief. Ender's commanders in the International Fleet isolate and manipulate him, conceal from him that he has killed two boys in fights, and deceive him into directing what he believes is a simulation and is in fact the battle that annihilates the Formic species. Nell is formed into a sovereign by devotion and founds a polity she knows to be real. Ishiguro's *Never Let Me Go* (2005) presents an education at Hailsham whose cultivation of sensibility is lavished on clones raised to have their vital organs removed until they die: the darkest version of the Mouse Army's prospect, in which refinement of the person and total instrumentalization of the body prove compatible. Chiang's and Ishiguro's later works, discussed in Chapter 7, complete the set.

9.2 *Fictions of abundance*

The genealogy of post-scarcity fiction begins in debate. Bellamy's *Looking Backward* (1888) imagined abundance through centralized administration; Morris's *News from Nowhere* (1890) answered with abundance through craft and fellowship; Wells's *A Modern Utopia* (1905) sought to synthesize them under a voluntary aristocracy. Samuel Johnson's *Rasselas* (1759) supplied the archetypal critique: the Happy Valley, where every want is met, is a prison from which the prince escapes in search of a 'choice of life'. The phyles are Rasselas's question institutionalized. Voltaire's *Candide* (1759) closed by enjoining the cultivation of one's garden, a counsel the Seed takes literally.

In the twentieth century, *Star Trek: The Next Generation* (1987–94) made the replicator a household object and declined to examine its sociology. Iain M. Banks's Culture novels (from 1987; theorized in 'A Few Notes on the Culture', 1994) imagined post-scarcity anarchism superintended by benevolent artificial Minds, so that formation is solved by superintelligent care. Cory Doctorow's *Down and Out in the Magic Kingdom* (2003) formalized positional goods as a reputation currency, 'Whuffie'; his *Walkaway* (2017) imagined defection from scarcity's institutions; Charles Stross's *Accelerando* (2005) imagined economics outrunning its human participants. Stephenson's dissent from all of them is consistent. Abundance does not produce a culture; cultures must produce people, and no Mind is available to do it for them.

9.3 *Polities of choice*

Le Guin's *The Dispossessed* (1974) is the indispensable comparison. Anarres, an anarchist society founded on voluntary association, has hardened into a conformity enforced not by law but by social conscience, and its physicist Shevek must leave in order to remain free. Le Guin diagnosed in anarchism the founder's paradox that Finkle-McGraw diagnoses in Victorianism: an order founded by dissidents tends to raise conformists. Delany's *Triton* (1976) imagined a heterotopia of chosen identities whose very freedom generates a new alienation. Ada Palmer's *Terra Ignota* sequence (2016–21) is the twenty-first century's most rigorous heir to the phyle. Its 'Hives' are non-territorial nations that adults choose; its 'Hiveless' are cousins of the thetes; and its child Bridger, who can bring toys to life, commands a small army of animated toy soldiers, a figure that rhymes uncannily with the Primer's toy companions and the Mouse Army. Palmer, a historian of early modern thought, writes in a pastiche of eighteenth-century narration as Stephenson writes in a pastiche of the nineteenth; both treat the formative repertoire of a past century as the operating system of the future. Bruce Sterling's *Islands in the Net* (1988), with its data havens and corporate polities, and *Schismatrix* (1985), with its rival posthuman clades, belong to the same lineage.

9.4 *Founders and daughters*

Octavia E. Butler's *Parable of the Sower* (1993) appeared two years before *The Diamond Age* and shares its deepest structure. In a collapsing America a young Black woman, Lauren Olamina, founds a community called Earthseed on a doctrine whose first principle is that God is Change. The coincidence of seed imagery is less striking than the coincidence of agency: in both novels the founder of the new order is a girl formed by catastrophe, and founding is an act of formation. Butler's seed is a doctrine; Stephenson's is a device. In *Parable of the Talents* (1998), Crusaders of the theocratic Christian America movement seize Earthseed's settlement, hold its adults as slaves in a 're-education' camp controlled by electronic collars, and take its children to be raised by Christian America families. That is the paideiarchic danger the Mouse Army embodies, stated without disguise: formation as conquest, beginning with the removal of children. Margaret Atwood's *The Handmaid's Tale* (1985) is Neo-Victorianism's dark twin: in Gilead the ruling Commanders hold fertile women as Handmaids in state-organized reproductive servitude that includes ritualized rape, where the Neo-Victorians rely on formation. Gilman's *Herland* (1915) is its utopian opposite.

The female *Bildungsroman*, anatomized in the collection edited by Elizabeth Abel, Marianne Hirsch, and Elizabeth Langland (1983), supplies the formal comparison. Charlotte Brontë's *Jane Eyre* (1847) is Nell's Victorian original: an abused orphan, a harsh school, a vocation as governess, an achieved autonomy. Jane's narrative closes in marriage, the conventional seal on female development. Nell's closes in the founding of a polity. The novel does not refuse the marriage plot so much as leave it behind.

9.5 *Stephenson's own arc*

Within Stephenson's work, *The Diamond Age* is a hinge. *Snow Crash* (1992) supplied the burbclaves, the metavirus, and the theory of language as executable code. *Cryptonomicon* (1999) and the Baroque Cycle (2003–04) turned backward to the origins of the modern system of money, science, and cryptography; the Cycle's Eliza, a girl taken into slavery who rises by intelligence to become a duchess and a financier, is Nell's ancestor written in reverse chronology. *Anathem* (2008) imagined 'concents', cloistered polities of theorists sustained by discipline and ritual: a mathic Neo-Victorianism. *Seveneves* (2015) made seven women the founders of seven peoples whose cultures descend from their characters. The thread running through all of them is the one this dissertation isolates: polities are formed by the persons who found them, and persons by the formation they receive. Stephenson's 'Innovation Starvation' (2011) and the anthology *Hieroglyph* (2014) argued that science fiction should supply engineers with motivating images. The Primer has become exactly such a hieroglyph, which is why the question of what it requires is no longer merely literary.

9.6 *The Parable of the Sower as latent schema*

A final comparison is speculative and structural. In the synoptic Parable of the Sower (Mark 4:3–20; Matthew 13:3–23), one seed falls on four grounds: the path, the stony ground, the thorns, and the good ground, which yields thirty, sixty, and a hundredfold. The novel sows one Primer in four grounds: an aristocrat's granddaughter, an engineer's daughter, a cohort of abandoned infants, and a thete child. The parable supplies the experimental design; the novel inverts its agronomy. The good ground, the soil of the hundredfold yield, is the stony ground of the thete's childhood, and the hundredfold yield is, almost literally, the Mouse Army. Butler drew her title from the same parable. The novels of 1993 and 1995 are twin parables of seed and founding.

CHAPTER

X

Coyote's Theorem

Toward a General Theory of Formative Sovereignty

THE PRECEDING CHAPTERS assemble the materials of an argument the novel conducts but never states. This chapter states it. Its claim is that the novel functions as a controlled thought experiment on the production of persons, and that its results can be expressed as four axioms and a theorem whose consequences extend beyond the fiction.

10.1 *Why formalize*

Formalization is warranted by the novel's own construction. Stephenson holds the apparatus constant and varies the conditions of its use. One design, the Primer, is deployed in four settings that differ along two dimensions: how far the child's world is ordered for her, and whether an individuated human voice inhabits the device. The outcomes diverge sharply. A design of this kind invites the reconstruction of the hypothesis it tests.

10.2 *Axioms*

Axiom 1 (Displaced Scarcity). Where the marginal cost of matter approaches zero, value concentrates in what cannot be compiled: positional goods, provenance, and formative attention (Chapter 2).

Axiom 2 (Phyletic Reproduction). In an order of universal exit and irrelevant territory, a polity persists only by producing persons who choose it; sovereignty is therefore exercised principally through formation (paideiarchy, Chapter 3).

Axiom 3 (Narrative Sovereignty). The authorship of the stories through which children come to love what they will later choose is the most fundamental form of rule available in such an order. This is Plato's premise in the *Republic* and Borges's in 'Tlön' (Chapter 6).

Axiom 4 (Compilation). An apparatus compiles when it produces outputs from a specification fixed in advance, such that any output is in principle derivable from the specification together with its inputs.

10.3 *The theorem*

Call a pupil *sovereign with respect to* a formative apparatus F when she can author a form of life that F's repertoire does not contain: not choose among F's options, but found an option F did not foresee.

> *Coyote's Theorem.* For any formative apparatus F that operates by compilation, sovereignty with respect to F cannot be an output of F alone. It arises only from the conjunction of F with (i) an *exteriority* E, conditions of the pupil's world that F does not script and could not script without converting them into lessons, and (ii) a *remainder* V, formative address given in excess of any contract and indexed to the pupil alone. Where either E or V is null, F yields competence, conformity, or exit, but not foundation.

Schematically, with Σ for sovereignty and R(F) for the repertoire of F:

$$\Sigma(p) \notin R(F), \qquad \Sigma(p) \neq \emptyset \;\Rightarrow\; E(p) \neq \emptyset \wedge V(p) \neq \emptyset$$

The proof is a diagonal argument in miniature. Suppose F could compile sovereignty. The novelty the sovereign introduces would then be derivable from F's specification and would belong to F's repertoire, contradicting the definition of sovereignty with respect to F. Suppose, more weakly, that F could predict which of its pupils will exceed it. The prediction would itself become part of the formation, and a pupil capable of sovereignty could act against it; the prediction defeats itself as the hypothetical halting-decider does in Turing's diagonal argument (1936). Gödel's incompleteness theorem (1931), that a sufficiently rich consistent system contains truths it cannot prove, has a pedagogical analogue: a sufficiently rich formative system admits capacities in its pupils that it cannot derive. Call this *formative undecidability*. Lady Lovelace's objection supplies the corollary in its original terms. A compiler can do whatever we know how to order it to perform. Sovereignty is precisely what we do not know how to order.

Why E and V, and nothing else? Because they are the two inputs compilation cannot absorb. Exteriority, once scripted, becomes curriculum: the Primer can teach Nell to survive danger, but the danger itself came from the Leased Territories, inflicted on her rather than offered to her. The remainder, once contracted, becomes service: the ractive company can sell hours of Miranda's time, but it cannot sell the increment that forms Nell. The theorem names the novel's discovery that formation has two irreducible exteriors, the world and the other person, and that a self capable of beginning something is formed where they intersect the code.

10.4 *Corollaries: the novel's experimental design*

Trajectory	Exteriority (E)	Remainder (V)	Outcome	Political form
Elizabeth Finkle-McGraw	Low: aristocratic shelter	Diffuse: no single devoted voice	Exit	Flight into CryptNet; subversion without foundation
Fiona Hackworth	Moderate: a father lost to the Drummers for a decade	Attenuated: a paternal voice reaching her through the Primer	Vocation	A life organized around answering a call
The Mouse Army	High: abandonment	Industrial: unindividuated	Allegiance	A people awaiting a sovereign
Nell	Maximal: beaten by her mother's boyfriends, no authority intervening; flight	Concentrated: Miranda's decade of devotion	Foundation	A new polity gathered around her

Corollary 1 (the Mouse Army). Formation with exteriority but without the remainder yields allegiance. The Mouse Army's girls are formed by hardship and by the Primer's code; they become competent, disciplined, and devoted, but their devotion seeks an object outside themselves. Formation at scale without individuated address produces a people, not a founder.

Corollary 2 (Elizabeth). Formation with code but without exteriority yields exit. Elizabeth receives the subversive curriculum in conditions that never test it and without the concentrated voice Nell receives. She vanishes from her family's estate, and her defection to CryptNet becomes a public sensation. Her grandfather takes it serenely: a young person who leaves, he tells Carl Hollywood, has outgrown credulity and will return once wise enough to see New Atlantis as the best of tribes. That reinterprets Hirschman's exit as an apprenticeship in loyalty, and the novel never shows the wager paying off. In Hirschman's terms she exits; in Arendt's she does not act.

Corollary 3 (Fiona). Formation with an attenuated remainder yields vocation, in the etymological sense of being called. Before his disappearance Hackworth told Fiona that her book was magic and that he would talk to her through it, and her mother reports that, to Fiona, he has never been gone. When her mother divorces him, Fiona defends him against every charge, and when he resurfaces she leaves with him for London. Her trajectory follows that voice rather than founding anything of its own. The remainder, even attenuated, determines direction.

Corollary 4 (Nell). Formation with maximal exteriority and a concentrated remainder yields foundation. Nell alone begins something. Exteriority must be named precisely here, because the theorem is easily misread. In Nell's case it took the form of beatings by the men in her mother's household and the absence of any authority to stop them: harm done to a child, which the theorem neither requires in that form nor recommends. The novel records what exteriority costs, in Harv's lungs, ruined by the residue the claves filter out, and in the abandonment from which the Mouse Army was gathered. What the theorem requires is only that formation not seal the child off from a world it does not script. 'Designed exteriority' in Section 10.7 means exposure to that world, never the infliction of adversity.

The table also exposes the novel's quiet refutation of its own Neo-Victorian frame. The phyle that commissioned the Primer could supply neither of the two inputs that made it effective. It shelters its children from exteriority by design, and it purchases its voices by contract. The commission could succeed only by being stolen.

10.5 *Natality, metamorphosis, trickster*

The theorem's outcome, foundation, is what Arendt called action. In *The Human Condition* (1958) action is the human capacity to begin, grounded in natality and therefore unpredictable in principle; in *On Revolution* (1963) its paradigmatic form is foundation, the establishment of a new political body. Nell's polity is an Arendtian foundation, and it arises, as Arendt would predict, from a capacity no institution could guarantee.

The novel gives natality a literal origin. Nell was conceived because her mother's contraceptive, a mite living in the womb that destroys eggs and can wear out without warning, had stopped working; the novel notes that Victorians disapproved of such devices, which could be bought from Chinese and Hindustani vendors. Nell is, in the strictest sense, the unscripted child, and the polity she founds is an unscripted people.

Nietzsche's 'On the Three Metamorphoses' in *Thus Spoke Zarathustra* (1883–85) supplies the developmental grammar. The spirit becomes a camel, bearing what is heaviest; then a lion, which speaks the sacred 'No' to the dragon called 'Thou shalt'; then a child, which is innocence, a new beginning, a sacred 'Yes'. Nell's trajectory follows the sequence. At the Academy she bears the Neo-Victorian law. In her passage beyond New Atlantis she refuses it. In her foundation she begins. Elizabeth, by contrast, reaches the lion and stops.

The theorem bears Coyote's name because the Primer's own design and romance know it. Explaining the device to Finkle-McGraw, Hackworth says that folklore consists of universal ideas mapped onto local cultures, and he takes the Trickster as his example: Coyote in the American Southwest, Raven on the Pacific coast, Reynard in Europe, Br'er Rabbit among African-Americans, and in twentieth-century literature first Bugs Bunny and then the Hacker. Finkle-McGraw remarks that in his youth the word hacker had a double meaning, one of them a trickster who broke into things. Paul Radin's *The Trickster* (1956), with Jung's commentary, and Lewis Hyde's *Trickster Makes This World* (1998) analyze the figure as the boundary-crosser who creates culture by breaking its rules. The Primer's romance ends at the Trickster's castle, and its final triumph coincides with the disclosure that King Coyote's world is run not by his machine but by persons, whereupon he yields it to Nell and charges her to make new ones (Section 6.6). A pedagogical romance that culminates in a trickster concedes that its summit is the rule-breaker, and that its last lesson is that it must be exceeded. Xunzi's indigo states the same truth in Confucian terms, Kant's paradox in Enlightenment terms, Finkle-McGraw's commission in Victorian ones; the engineer who encoded the Trickster as a universal is named Hackworth. Coyote's Theorem states why the truth holds.

10.6 *The ethnogenetic Bildungsroman*

Bakhtin distinguished, among the types of *Bildungsroman*, a highest form in which the hero emerges along with the world, so that personal formation and historical transformation coincide. Franco Moretti's *The Way of the World* (1987) read the classical *Bildungsroman* as the symbolic form of modernity, reconciling individual autonomy with socialization. Nell's story reconciles nothing. It neither integrates the heroine into an existing order, as Goethe's apprenticeship novel does, nor records her defeat by one, as the nineteenth-century novel of disillusionment does. It ends with the founding of an order around her.

I propose the term *ethnogenetic Bildungsroman* for a narrative in which the formation of a person and the genesis of a people are one event. Its archetypes lie outside the novel. Moses, raised in Pharaoh's household, leads out a people. Virgil's Aeneas carries a city in his person. The nearest is Joan of Arc, a peasant girl guided by voices who commanded an army, a figure retold by Mark Twain (1896) and George Bernard Shaw (1923). Nell is a Joan whose voice is a ractor's and whose army was raised by a book. The Mouse Army is the Primer's diegesis made polity, Tlön entering the world: the romance of Princess Nell acquires subjects. Bakhtin's emergence with the world here becomes literal.

The novel names the outcome in nearly these terms. Having watched the girls in the streets of Pudong, Carl Hollywood concludes that the Mouse Army constitutes a new ethnic group of sorts, with Nell its undisputed leader, and the Neo-Victorian sovereign, Victoria, accords Nell equivalent status as a fellow sovereign. Nell's first act in that role is to accept the army's allegiance and direct it toward the thousands of stranded travellers from New Atlantis, Nippon, Israel, and the other Outer Tribes. The ethnogenetic Bildungsroman ends with a people, a sovereign, and an act of protection.

10.7 *Normative coda: the Primer after 2023*

When the novel appeared, the Primer was a fantasy. Since 2023, large language model tutors have realized much of its apparatus at negligible marginal cost: adaptive narrative, Socratic questioning, unlimited patience, multimodal voice. The Primer has become a recurrent touchstone in the rhetoric of AI-tutoring ventures, and Bloom's two sigmas are again invoked as within reach. Coyote's Theorem yields three predictions and three proposals.

The predictions follow from the corollaries. First, machine tutoring delivered at scale without an individuated human remainder will reproduce the Mouse Army's outcome: real competence and a diffuse allegiance, to a platform, a provider, or a narrative, rather than sovereignty. Second, the principal danger is not incompetence but homogeneity. Hundreds of millions of children formed in part by a handful of models constitute a paideiarchy by default, wielded, under Axiom 3, by whoever authors those models' dispositions. Third, the children best served will be those whose machine tutors are embedded in rich exteriority and sustained human regard, so that the advantage will accrue to those who already possess both. Formative attention will be the positional good of the coming decades.

The proposals follow from the remainder's definition.

1. *A right to an uncompilable interlocutor.* Every child formed in part by machine pedagogy has a claim to sustained, individuated human address. Love cannot be mandated, but its conditions can be guaranteed institutionally: continuity of persons, time, small ratios, and protection of the adults who provide them from the cost disease that makes their labour appear expensive.
2. *Formative pluralism.* Narrative sovereignty should be treated as a concentration risk. No single provider should author the formative stories of a population, for the reason Plato understood and the Mouse Army demonstrates.
3. *Designed exteriority.* Formative systems should be built to hand their pupils to the world rather than to retain them. The novel ends by turning the Primer's reader outward, toward the person whose voice animated it. A tutor that maximizes its own use fails the theorem's first condition.

One objection must be met. If machines come to care, in the sense Ishiguro's Klara leaves open, the remainder need no longer be human. The theorem is indifferent to the substrate. It specifies properties, not species: time-indexed, person-indexed, non-contractual in its decisive increment, reciprocally binding. A system that possessed all four would supply V. Such a system would also, by the fourth property, be bound to its pupil in ways no provider could own. Whether it can be built is an engineering question. Whether it could be sold is the Second Paradox of Value.

CONCLUSION

Drowning the Book

The Diamond Age ends abruptly, as Stephenson's novels habitually do, and the abruptness has often been read as a failure of architecture. Read through the argument of this dissertation, it is the novel's final formal act. The book closes at the moment its heroine meets, without mediation, the woman whose voice raised her. At that moment the Primer has nothing further to do, and neither has the novel. A book can represent compilation in exhaustive detail and can narrate the remainder's effects; it cannot represent the remainder itself, the unscripted presence of one person to another, without converting it into text and thus into a kind of specification. The novel performs its own obsolescence. Like the Primer, it ends where the book is no longer needed.

The argument may now be gathered. Stephenson abolishes material scarcity in order to observe what remains scarce, and finds that scarcity has migrated to the formation of persons. In a world of elective, post-territorial polities, formation becomes the decisive instrument of sovereignty, and the Neo-Victorian and Confucian restorations emerge as rival technologies for reproducing virtue without scarcity's discipline. The novel's central artifact proves to be a machine whose formative power resides in the human being concealed within it, and whose code, reproduced at scale, yields allegiance rather than autonomy. Its submerged collective spends the surplus of abundance in the dissolution of persons. Its experimental design yields a theorem: no apparatus of formation can compile the capacity that would exceed it, because that capacity arises only where code meets an unscripted world and an unpurchased voice.

Smith's diamond was valuable because it was scarce and nearly useless. The diamond of the diamond age is valuable because it is indispensable and cannot be produced on demand: the regard of one person for another, given beyond what is owed. Every institution in the novel is a strategy for capturing that regard, industrializing it, or doing without it, and each fails in proportion to its success at compilation. Prospero drowns his book and returns to Milan to rule. Nell descends past the book, retrieves the voice, and founds. Candide's counsel was to cultivate one's garden, and the Seed makes the counsel literal for matter. The novel's deeper counsel concerns persons: the garden in which a self grows is tended, finally, by someone, and no age of diamonds alters the arithmetic of that tending.

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The Diamond Age; Neal Stephenson; speculative fiction; philosophical fiction; postcyberpunk; nanotechnology; molecular manufacturing; matter compiler; post-scarcity economics; paradox of value; positional goods; attention economy; formative scarcity; uncompilable remainder; paideia; paideiarchy; philosophy of education; Bildungsroman; ethnogenetic Bildungsroman; phyle; elective community; exit and voice; statelessness; network state; Neo-Victorianism; hypocrisy; Confucianism; techno-Orientalism; emotional labour; social reproduction; Mechanical Turk; artificial intelligence in education; natality; trickster; Coyote's Theorem